

Solutions

Maths

Q1. If z_1, z_2, z_3 are the solutions of $z^2 + \bar{z} = z$, then $z_1 + z_2 + z_3$ is equal to (where z is a complex number on the argand plane & $i = \sqrt{-1}$) :

- (A) $2 + 2i$ (B) $2 - 2i$ (C) 0 (D) 2

Ans. D

Sol. $z^2 + \bar{z} = z$ _____ (i)

Taking conjugate

$\bar{z}^2 + z = \bar{z}$ _____ (ii)

adding (i) & (ii)

$$z^2 + \bar{z}^2 = 0$$

$$\Rightarrow \bar{z} = iz \text{ or } \bar{z} = -iz$$

Case-I : $\bar{z} = iz$

$\Rightarrow$ from (i)

$$z^2 + iz = z$$

$$\Rightarrow z = 0, 1 - i$$

Case-II : $\bar{z} = -iz$

$$\Rightarrow z^2 - iz = z$$

$$\Rightarrow z = 0, 1 + i$$

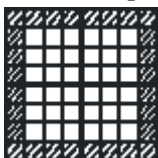
$$\Rightarrow z_1 + z_2 + z_3 = 2$$

Q2. Number of squares that can be formed in 8×8 chessboard, so that each square is surrounded by four squares of same dimension :

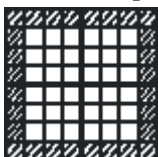
- (A) 36 (B) 42 (C) 45 (D) 48

Ans. C

Sol. Case I : Squares of 1×1 dimension $= 6 \times 6 = 36$



Case II : square of 2×2 dimension $= 3 \times 3 = 9$



No other possibility is there hence total squares $= 36 + 9 = 45$

Q3. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$, how many one - one function $f : A \rightarrow B$ can be define so that $f(i) \neq 2i - 1, \forall i = 2, 3, 4, 5$:

- (A) 265 (B) 53 (C) 88 (D) None of These

Ans. B

Sol. $D_4 + D_5 = 53$

Q4. In how many ways 4 square can be chosen on a chessboard, such that all the squares lie in a diagonal :

- (A) 504 (B) 182 (C) 364 (D) 252

Ans. C

Sol. Let us consider the $\triangle ABC$. Number of ways in which 4 selected squares are along the lines $A_4C_4, A_3C_3, A_2C_2, A_1C_1$ and AC are ${}^4C_4, {}^5C_4, {}^6C_4$ and 8C_4 respectively.

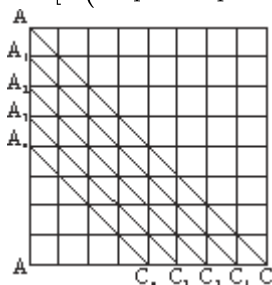
Similarly, in $\triangle ACB$, number of ways in which 4 selected squares are along the diagonal line parallel to AC are ${}^4C_4, {}^5C_4, {}^6C_4, {}^7C_4$ and 8C_4 , but 8C_4 triangles occur only once.

Hence the total number of ways in which the 4 selected squares are in a diagonal line parallel to AC are

$$2[2({}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4) + {}^8C_4]$$

Also same is the case of selecting 4 squares on a chessboard. Such that the 4 squares are in a diagonal line

$$= 2[2({}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4) + {}^8C_4]$$



Q5. Let the product of all the divisors of 1440 be P. If P is divisible by 24^x , then the maximum value of x is :

- (A) 28 (B) 30 (C) 32 (D) 36

Ans. B

Sol. $1440 = 2^5 \cdot 3^2 \cdot 5^1$

$$\text{No. of divisors} = (5 + 1) \cdot (2 + 1) \cdot (1 + 1) = 36$$

Product of divisors = $1 \cdot 2 \cdot 3 \dots 480 \cdot 720 \cdot 1440$. Here all the 36 divisors are written in the increasing order. They can be clubbed into 18 pairs, as shown below.

$(1 \cdot 1440) \cdot (2 \cdot 720) \cdot (3 \cdot 480) \dots$ etc.

$\therefore$ Product of divisors $(1440)^{18} = 290.336.518 = (23 \cdot 3)30.36.518 = 2430.36.518$ which is divisible by 24^x

$\therefore$ Maximum value of x = 30

Q6. Let $n_1 = x_1x_2x_3$ and $n_2 = y_1y_2y_3$ be two 3-digit numbers, then the pairs of n_1 and n_2 can be formed so that n_1 can be subtracted from n_2 without borrowing, is :

- (A) $55.(54)$ (B) $45.(55)$ (C) $45.(55)^2$ (D) $55.(45)^2$

Ans. C

Sol. Given $n_1 = x_1x_2x_3$ and $n_2 = y_1y_2y_3$

n_1 and n_2 can be subtracted without borrowing at any stage, if $x_i \leq y_i$

Value of x_i	Value of y_i
9	0, 1, 2,, 9
8	0, 1, 2,, 8
7	0, 1, 2,, 7
6	0, 1, 2,, 6
5	0, 1, 2,, 5
4	0, 1, 2,, 4
3	0, 1, 2, 3
2	0, 1, 2
1	0, 1
0	0

Thus, x_3 and y_3 can be selected collectively by $10 + 9 + 8 + \dots + 1 = 55$ ways. Similarly, (x_2, y_2) can be selected in 55 ways. But pair (x_1, y_1) can be selected in $1 + 2 + 3 + \dots + 9 = 45$ ways as in this pair we cannot have zero.

Thus, total number of ways in $45.(55)^2$

Q7. The value of $\sum_{r=1}^5 {}^{20}C_{2r-1}$ is :

- (A) 2^{18} (B) 2^{19} (C) 2^{20} (D) $2^{19} - {}^{20}C_{10}$

Ans. A

Sol. $\sum_{r=1}^5 {}^{20}C_{2r-1} = {}^{20}C_1 + {}^{20}C_3 + \dots + {}^{20}C_9$
 $= \frac{1}{2} [{}^{20}C_1 + {}^{20}C_3 + \dots + {}^{20}C_9 + {}^{20}C_{11} + \dots + {}^{20}C_{19}]$
 $= \frac{1}{2} [2^{19}] = 2^{18}$

Q8. Greatest binomial coefficient in the expansion of $(1+x)^{40+m}$ is $\sum_{i=0}^m {}^m C_i {}^{40}C_{m-i}$, then the value of m is (where m is an even natural number) :

- (A) 10 (B) 20 (C) 30 (D) 40

Ans. D

Sol. Middle term has greatest binomial coefficient $\therefore {}^{40+m}C_{\frac{20+m}{2}} = \sum_{i=0}^m {}^m C_i {}^{40}C_{m-i}$
 ${}^{40+m}C_{\frac{20+m}{2}} = {}^{40+m}C_m \therefore 20 + \frac{m}{2} = m \Rightarrow m = 40$

Q9. Remainder when 7^{636} is divided by 19 is equal to :

- (A) 1 (B) 7 (C) 12 (D) 13

Ans. A

Sol. $7^{636} = (343)^{212} = (1 + 19 \times 18)^{212}$
 $= 1 + {}^{212}C_1(19 \times 18) + \dots$
 $\Rightarrow \text{Remainder} = 1$

Q10. The value of ${}^{505}C_5 - 5 \cdot {}^{404}C_5 + 10 \cdot {}^{303}C_5 - 10 \cdot {}^{202}C_5 + 5 \cdot {}^{101}C_5$ is equal to :

- ${}^{505}C_5 - 5 \cdot {}^{404}C_5 + 10 \cdot {}^{303}C_5 - 10 \cdot {}^{202}C_5 + 5 \cdot {}^{101}C_5$
 (A) $({}^{101}C_1)^5$ (B) ${}^{101}C_1$ (C) $({}^{201}C_1)^5$ (D) ${}^{201}C_5$

Ans. A

Sol. ${}^{505}C_5 - 5 \cdot {}^{404}C_5 + 10 \cdot {}^{303}C_5 - 10 \cdot {}^{202}C_5 + 5 \cdot {}^{101}C_5 - 1 + 1$
 Coefficient of x^5 in
 $(1+x)^{505} - 5(1+x)^{404} + 10(1+x)^{303} - 10(1+x)^{202} + 5(1+x)^{101} - 1 + 1$
 $= \text{coefficient of } x^5 \text{ in } [1+x]^{101} - 1]^5 + 1$
 $= ({}^{101}C_1)^5$

Q11. Two fair dice are rolled. The probability that the first time sum of numbers will appear on them is 6 occurs at odd throw is $\frac{a}{b}$, then the value of $(b-a)$ is (where a and b are coprime numbers) :

- (A) 28 (B) 31 (C) 21 (D) 23

Ans. B

Sol. Favourable cases for sum 6 is 5

$\therefore$ required probability

$$\frac{5}{36} + \left(\frac{31}{36}\right)^2 \times \frac{5}{36} + \left(\frac{31}{36}\right)^4 \times \frac{5}{36} + \dots = \frac{36}{67}$$

Q12. Let a, b, c $\in \mathbb{I}^+$ such LCM of a and b is 1000, LCM of b and c is 2000 and LCM of c and a is 2000, then number of ordered triplet (a, b, c) is :

- 63
 (A) 63 (B) 90 (C) 70 (D) None of these

Ans. C

Sol. We have, $1000 = 2^3 \cdot 5^3$, $2000 = 2^4 \cdot 5^3$
 So, $a = 2^A \cdot 5^R$, $b = 2^B \cdot 5^S$, $c = 2^C \cdot 5^T$

$$\max(A, B) = 3, \max(A, C) = \max(B, C) = 4$$

$$\max(R, S) = 3, \max(R, T) = \max(S, T) = 3$$

There are 10 ways of choosing R, S, T.

1 with all 3, 3 with one 2, 3 with one 1, 3 with one 0, C must be 4.

There are 7 ways of choosing A, B.

1 with both 3, 3 with $A = 3$, B not, 3 with $B = 3$ and A not.

Thus, 7 ways of choosing A, B, C have $7 \cdot 10 = 70$ ways

Q13. The coefficient of x^4 in the expansion of $(e^x - 1)^2$ is :

(A) $\frac{3}{8}$ (B) $\frac{5}{12}$ (C) $\frac{7}{12}$ (D) $\frac{5}{24}$

Ans. C

Sol.
$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}$$
$$e^{2x} = 1 + \frac{2x}{1!} + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!}$$
$$\text{co-efficient of } x^4 = \frac{2^4}{4!} - \frac{2}{4!} = \frac{7}{12}$$

Q14. $\sum_{0 \leq i < j \leq n} (i+j)(C_i^2 + C_j^2 + C_i C_j)$ is equal to :

(A) $\frac{n}{2} \{(2n-1)^{2n} C_n + 4^n\}$ (B) $\frac{n}{2} \{(2n+1)^{2n} C_n + 4^n\}$ (C) $\frac{n}{2} \{4^n - (2n-1) \cdot^{2n} C_n\}$
(D) $\frac{n}{2} \{4^n - (2n+1) \cdot^{2n} C_n\}$

Ans. A

Sol. Let $S = \sum \sum (i+j)(C_i^2 + C_j^2 + C_i C_j)$
Replacing i by n-i and j by n-j
$$S = \sum \sum (n-i+n-j)(C_{n-i}^2 + C_{n-j}^2 + C_{n-i} \cdot C_{n-j})$$
$$\Rightarrow 2S = 2n \sum \sum (C_i^2 + C_j^2 + C_i C_j)$$
$$S = n \left\{ n \cdot^{2n} C_n + \frac{4^n -^{2n} C_n}{2} \right\}$$
$$= n \frac{\{(2n-1)^{2n} C_n + 4^n\}}{2}$$

Q15. The value of the expression

$$1 - \frac{n}{1} \cdot \frac{(1+x)}{1+nx} + \frac{n(n-1)}{1 \cdot 2} \frac{(1+2x)}{(1+nx)^2} - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} \frac{(1+3x)}{(1+nx)^3} + \dots \dots (n > 0) \text{ is:}$$

(A) 2 (B) 1 (C) 3 (D) 0

Ans. D

Sol. The expression can be divided into two parts.

$$\left(1 - \frac{n}{1} \frac{1}{(1+nx)} + \frac{n(n-1)}{2} \frac{1}{(1+nx)^2} - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} \frac{1}{(1+nx)^3} + \dots \right)$$

$$+ \left(\frac{-nx}{1+nx} + \frac{n(n-1)x}{(1+nx)^2} - \frac{n(n-1)(n-2)x}{1 \cdot 2 \cdot 3(1+nx)^3} + \dots \right) \\
\left(1 - \frac{1}{1+nx} \right)^n - \frac{nx}{(1+nx)} \left(1 - \frac{n-1}{1(1+nx)} + \frac{(n-1)(n-2)}{1 \cdot 2(1+nx)^2} + \dots \right) \\
\left(\frac{nx}{1+nx} \right)^n - \frac{nx}{(1+nx)} \left(1 - \frac{1}{1+nx} \right)^{n-1} \left(\frac{nx}{1+nx} \right)^n - \left(\frac{nx}{(1+nx)} \right) \left(\frac{nx}{(1+nx)} \right)^{n-1} = 0$$

Q16. Let z_1, z_2, z_3 are roots of the equation $z^3 + z^2(i-1) + z(2-i) - 2 = 0$, then $\text{Re}(z_1) + \text{Re}(z_2) + \text{Re}(z_3)$ is :

(A) -1 (B) 0 (C) 1 (D) 2

Ans. C

Sol. $(z-1)(z-i)(z+2i) = 0$

$$z_1 = 1, z_2 = i, z_3 = -2i$$

$$\text{Re}(z_1) + \text{Re}(z_2) + \text{Re}(z_3) = 1$$

Q17. If $P = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$, $a = P + P^2 + P^4$ and $b = P^3 + P^5 + P^6$, then the value of $|(a+b) + abi|$ (where $i = \sqrt{-1}$) is :

(A) $\sqrt{3}$ (B) $\sqrt{5}$ (C) $\sqrt{6}$ (D) 2

Ans. B

Sol. $P = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$

$$P = e^{i\frac{2\pi}{7}}$$

$$\Rightarrow P^7$$

$$= \left(e^{i\frac{2\pi}{7}} \right)^7$$

$$= e^{2\pi i}$$

$$a + b = P + P^2 + P^3 + P^4 + P^5 + P^6$$

$$a + b + 1 = 1 + P + P^2 + \dots + P^6$$

$$= \frac{P^7 - 1}{P - 1} \{P^7 = 1\}$$

$$a + b + 1 = 0$$

$$\therefore a + b = -1$$

$$ab = (P + P^2 + P^4)(P^3 + P^5 + P^6)$$

$$= P^4 + P^6 + P^7 + P^5 + P^7 + P^8 + P^7 + P^9 + P^{10}$$

$$= P^4 + P^6 + P^5 + P^8 + P^9 + P^{10} + 3$$

$$P^4 + P^5 + P^6 + P^7(P + P^2 + P^3) + 3$$

$$P + P^2 + \dots + P^6 + 3$$

$$-1 + 3 = 2$$

$$\therefore |(a+b) + abi|$$

$$= |-1 + 2i| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

Q18. If $\alpha_1, \alpha_2, \dots, \alpha_n$ are the n^{th} roots of unity then ${}^nC_1\alpha_1 + {}^nC_2\alpha_2 + \dots + {}^nC_n\alpha_n$ is equal to :

(A) $\left(1 + \frac{\alpha_2}{\alpha_1}\right)^n - 1$ (B) $\frac{\alpha_1}{2}[(1 + \alpha_1)^n - 1]$ (C) $\frac{\alpha_1 + \alpha_{n-1}}{2}$ (D) $(\alpha_1 + \alpha_{n-1})^n - 1$

Ans. A

Sol. ${}^nC_1 \cdot \alpha + {}^nC_2 \cdot \alpha^2 + {}^nC_3 \cdot \alpha^3 + \dots + {}^nC_n \cdot \alpha^n = (1 + \alpha)^n - 1$
 where $\alpha = e^{n \frac{2\pi i}{\alpha_1}}$

Q19. If the letters of the word SIKAR are arranged in all possible ways and these words are written out as in dictionary, then the word SIKAR appears at serial number k and Three distinct numbers a, b, c are selected at random from the set of first 100 natural numbers. The probability that a, b, c are in A.P. is given by $\frac{m}{n}$. Where m, n are relatively prime natural number, then the last digit of $(m + n + k)$ is :

(A) 7 (B) 2 (C) 5 (D) 1

Ans. B

Sol. AIKRS

A = 24

I = 24

K = 24

R = 24

SA = 6

SIA = 2

SIKAR = 1

SIKAR = 105th rank

a, b, c are in A.P.

$\Rightarrow a + c = 2b$

$\Rightarrow a + c$ is even

$\Rightarrow a$ and c both are even or both are odd

$\therefore P = \frac{{}^{50}C_2 + {}^{50}C_2}{{}^{100}C_3} = \frac{1}{66}$

$\therefore m + n = 67$

Q20. If z is a complex number satisfying the equation $z^6 - 6z^3 + 25 = 0$, then the value of $|z|$ is

(A) $5^{1/3}$ (B) $25^{1/3}$ (C) $125^{1/3}$ (D) $625^{1/3}$

Ans. A

Sol. Let $z^3 = t$

$\Rightarrow t^2 - 6t + 25 = 0$

$\Rightarrow t = \frac{6 \pm \sqrt{-64}}{2} = 3 \pm 4i$

$\Rightarrow z = (3 \pm 4i)^{1/3}$

$\Rightarrow |z| = |3 \pm 4i|^{1/3} = 5^{1/3}$

Q21. If z is a complex number such that $z + i|z| = \bar{z} + 1$, then $|z|$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Ans. C

Sol. Put $z = x + iy$

$$(x + iy) + i(|z|) = i(x - iy) + 1$$

$$\Rightarrow x = y + 1 \text{ \& } y + |z| = x$$

$$\Rightarrow x - y = 1 \text{ \& } x - y = |z|$$

$$\Rightarrow |z| = 1$$

Q22. Let C_1 and C_2 be two curves on the complex plane defined as

$$C_1 : z + \bar{z} = 2(|z| - 1)$$

$$C_2 : \arg(z + 1 + i) = \alpha$$

where $\alpha \in (0, \pi)$ such that the curves C_1 and C_2 touches each other at P (z_0) then the value of $|z_0|^2$ is :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $C_1 : z + \bar{z} = 2|z| - 1$

$$\text{Put } z = x + iy$$

$$2x = 2|x - 1 + iy|$$

$$x^2 = (x - 1)^2 + y^2$$

$$Y^2 = 2\left(x - \frac{1}{2}\right)$$

$$C : \arg(z - (-1 - i)) = \alpha$$

It's a ray emanating from $(-1, -1)$ and making angle α with the positive real axis C_1 and C_2 have exactly one point common

C_2 must be tangent to C_1

$$\therefore y = m\left(x - \frac{1}{2}\right) + \frac{1}{2m} \text{ passes to } (-1, -1)$$

$$\Rightarrow m = 1, m = -\frac{1}{3}$$

for $m = -\frac{1}{3}$ the ray will not be tangent

$$(-1, -1)$$

$y = x$ is tangent to the parabola

$$P(Z_0) = 1 + i$$

$$|Z_0| = \sqrt{2}$$

Q23. If $z = re^{i\theta}$ ($r > 0$ & $0 \leq \theta < 2\pi$) is a root of the equation $z^8 - z^7 + z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$ then number of values of ' θ ' :

(A) 6 (B) 7 (C) 8 (D) 9

Ans. C

$$\text{Sol. } \frac{(1 + z^9)}{1 + z} = 0, z \neq -1$$

$$\Rightarrow z^9 = -1$$

$$\Rightarrow re^{i\theta} = e^{\frac{i(2n+1)\pi}{9}}, n = 1, 2, \dots, 8$$

- Q24. If z_1, z_2, z_3 are complex number such that $|z_1| = 2, |z_2| = 3, |z_3| = 4$ then maximum value of $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ is :
- (A) 58 (B) 29 (C) 87 (D) None of these

Ans. C

Sol. $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$
 $= 2(4 + 9 + 16) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
 where $\vec{a}, \vec{b}, \vec{c}$ are position vectors of points z_1, z_2, z_3
 $\Rightarrow \text{Maximum value} = 58 - 2 \times \left(-\frac{29}{2}\right) = 87$

- Q25. Value of $\sum_{k=1}^{100} (i^{k!} + \omega^{k!})$, where $i = \sqrt{-1}$ and ω is complex cube root of unity, is :
- (A) $190 + \omega$ (B) $192 + \omega^2$ (C) $190 + i$ (D) $192 + i$

Ans. D

Sol. $\sum_{k=1}^{100} i^{k!} + \sum_{k=1}^{100} \omega^{k!}$
 $\sum_{k=1}^{100} i^{k!} = i^{1!} + i^{2!} + i^{3!} + i^{4!} + \dots + i^{100!}$
 $= i - 1 + i^6 + 1 + 1 + \dots + 1 = i - 2 + 97$
 $= i + 95.$
 $\sum_{k=1}^{100} \omega^{k!} = \omega^{1!} + \omega^{2!} + \omega^{3!} + \omega^{4!} + \dots + \omega^{100!}$
 $= \omega + \omega^2 + 1 + 1 + 1 + \dots + 1 = 97$
 sum $= i + 95 + 97 = i + 192$

- Q26. Three different numbers are selected at random from the set $A = \{1, 2, 3, \dots, 10\}$. Then the probability that the product of two numbers equal to the third is $\frac{p}{q}$, where p and q are relatively prime positive integers then the value of $(p + q)$:
- (A) 39 (B) 40 (C) 41 (D) 42

Ans. C

Sol. If product of two numbers equal to third number, then possibilities are (2, 3, 6) (2, 4, 8) (2, 5, 10)
 Probability $= \frac{3}{10C_3} = \frac{1}{40}$

- Q27. To obtain a gold coin; 6 men, all of different weight, are trying to build a human pyramid as shown in the figure. Human pyramid is called "stable" if some one not in the bottom row is "supported by" each of the two closest people beneath him and no body can be supported by anybody of lower weight. Formation of 'stable' pyramid is the only condition to get a gold coin. What is the probability they get gold coin
- (A) $\frac{1}{45}$ (B) $\frac{2}{45}$ (C) $\frac{4}{45}$ (D) $\frac{1}{5}$

Ans. A

Sol. Total case = $n = 6!$

Favourable case = $m = (3! \times 2!) + (2! \times 2!) = 16$

$$\text{Probability} = \frac{16}{6!} = \frac{1}{45}$$

Q28. Given z is a complex number satisfying $z^2 - z - |z|^2 + \frac{64}{|z|^5} = 0$ and $\text{Re}(z) \neq \frac{1}{2}$ (where $\text{Re}(z)$ denotes real part of z), then $|z|$ is less than :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C, D

Sol. $\therefore z^2 - z = |z|^2 - \frac{64}{|z|^5}$ = a purely real number

$$\Rightarrow z^2 - z = \overline{z^2 - z}$$

$$\Rightarrow (z^2 - z) - (\overline{z^2 - z}) = 0$$

$$\text{if } z - \bar{z} = 0 \Rightarrow z = \bar{z}$$

$\Rightarrow z$ is purely real number

So put $z = x$ in given equations

$$x^2 - x - |x|^2 + \frac{64}{|x|^5} = 0$$

$$\Rightarrow x^2 - x - |x|^2 + \frac{64}{x^5} = 0 \text{ (if } x > 0 \text{)}$$

($\therefore x < 0$ not possible)

$$\Rightarrow x^6 = 64$$

$$\Rightarrow x = 2$$

$$\therefore |z| = 2$$

Q29. If $x + \frac{1}{x} = 1$ and $a = x^{2010} + \frac{1}{x^{2010}}$ and b be the digit at unit place in the number $2^{2n} + 1$, $n \in N$ and $n > 1$, then $a + b$ can be equal to :

(A) 4 (B) 6 (C) 7 (D) 9

Ans. C, D

Sol. $x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0$

$$\Rightarrow x = \frac{1}{2} \pm i \frac{\sqrt{3}}{2} = \cos \frac{\pi}{3} \pm i \sin \frac{\pi}{3}$$

$$\Rightarrow x^{2010} = \cos \frac{2010\pi}{3} \pm i \sin \frac{2010\pi}{3} = 1$$

$$\therefore a = x^{2010} + \frac{1}{x^{2010}} = 2$$

$$b = \text{unit place digit of } 4^n + 1 \Rightarrow b = 5 \text{ or } b = 7$$

Q30. Let set $A = \{1, 2, 3, \dots, 22\}$. Set B is a subset of A and B has exactly 11 elements. The sum of elements of all possible sets B is :

(A) divisible by 8 (B) divisible by 4 (C) divisible by 11 (D) divisible by 17

Ans. B, C, D

Sol. Sum = $1(^{21}C_{10}) + 2(^{21}C_{10}) + \dots + 22(^{21}C_{10})$

Q31. The no. of integers from 1 to 10^5 which contain exactly one 3, exactly one 4 and exactly one 5 must be :

(A) more than 2000 (B) more than 3000 (C) 2940 (D) 3270

Ans. A, C

Sol. Required number is $5 \times 4 \times 3 \times 7 \times 7 = 2940$

Q32. If $\sum_{r=0}^n \frac{r}{^nC_r} = \sum_{r=0}^n \frac{n^2 - 4n + 6}{2 \cdot ^nC_r}$, then :

(A) $n = 1$ (B) $n = 2$ (C) $n = 3$ (D) $n = 4$

Ans. B, C

Sol.
$$\sum_{r=0}^n \frac{r}{^nC_r} = \sum_{r=0}^n \frac{n-r}{^nC_{n-r}}$$
$$= \frac{n}{2} \sum_{r=0}^n \frac{1}{^nC_r}$$
$$\Rightarrow \frac{n}{2} = \frac{n^2 - 4n + 6}{2}$$
$$\Rightarrow n^2 - 4n + 6$$
$$\Rightarrow n^2 - 5n + 6 = 0$$
$$\Rightarrow n = 2, 3$$

Q33. If 3 numbers are chosen randomly from the set $\{1, 2, 2^2, \dots, 2^n\}$ without replacement then the probability they are in increasing geometric progression is :

(A) $\frac{3}{2n}$ (n odd) (B) $\frac{4}{3n-1}$ (n odd) (C) $\frac{3n}{2(n^2-1)}$ (n even) (D) $\frac{2n}{n^2+4}$ (n even)

Ans. A, C

Sol. Case-I n is even $n = 2$

$$\text{Required probability} = \frac{{}^{k+1}C_2 + {}^kC_2}{{}^{2k+1}C_3} = \frac{3n}{2(n^2-1)}$$

Case-II. n : odd, $n = 2k + 1$

$$\text{Required probability} = \frac{{}^{2k+1}C_2 - C_2}{{}^{2k+2}C_3} = \frac{3}{2n}$$

Q34. Consider the system of equations

$$px^2 + qx = 0$$

$$rx^2 + sx = 0 \text{ where } p, q, r, s \in \{2, 3, 4, 6\}$$

Statement I : The probability that the system of equations has two solutions is $\frac{9}{64}$.

Statement II : It is given that the system of equations has exactly one solution, then the probability that

$$p = r \text{ is } \frac{12}{55}$$

Then the correct option(s) is/are :

- (A) Statement I is true (B) Statement II is true (C) Statement I is false
(D) Statement II is false

Ans. A, B

Sol. St-1: $\frac{p}{r} = \frac{q}{s}$
 $\frac{p}{r} = 1 = \frac{q}{s} \Rightarrow 16 \text{ pairs}$
 $\frac{p}{r} = \frac{2}{3} \text{ or } \frac{p}{r} = \frac{3}{2} \Rightarrow 8 \text{ pair}$
 $\frac{p}{r} = \frac{1}{2} \text{ or } \frac{p}{r} = 2 \Rightarrow 8 \text{ pair}$
 $\frac{p}{r} = \frac{1}{3} \text{ or } \frac{p}{r} = 3 \Rightarrow 2 \text{ pair}$
 $\frac{p}{r} = \frac{3}{4} \text{ or } \frac{p}{r} = \frac{4}{3} \Rightarrow 2 \text{ pair}$

$$\text{Probability} = \frac{36}{4 \times 4 \times 4} = \frac{9}{64}$$

 St-2 : $p = r$, then $q \neq s$ total favourable = 12×4

$$\text{Probability} = \frac{12 \times 4}{220} = \frac{12}{55}$$

Q35. A bag contains four tickets marked with 112, 121, 211, 222 one ticket is drawn at random from the bag. Let E_i ($i = 1, 2, 3$) denote the event that i^{th} digit on the ticket is 2. Then :

- (A) E_1 and E_2 are independent (B) E_2 and E_3 are independent (C) E_3 and E_1 are independent
(D) E_1, E_2, E_3 are independent

Ans. A, B, C

Sol. We have $P(E_i) = 2/4 = 1/2$ or $i = 1, 2, 3$. Also for $i \neq j$, $P(E_i \cap E_j) = 1/4 = P(E_i)P(E_j)$.
 Therefore, E_i and E_j are independent for $i \neq j$.
 Also, $P(E_1 \cap E_2 \cap E_3) = 1/4 \neq P(E_1)P(E_2)P(E_3)$
 $\therefore E_1, E_2, E_3$ are not independent.

Q36. Let P_k be a regular polygon of k sides each of whose sides has length unity. Consider set $A = \{P_3, P_4, P_5, \dots, P_{10}\}$. An element is chosen from set A; then :

- (A) Probability that all diagonals of the chosen polygon are equal in length is $\frac{1}{8}$.
 (B) Probability that all diagonals of the chosen polygon equal in length is $\frac{1}{4}$.
 (C) Probability that number of diagonals of chosen polygon is an odd number is $\frac{1}{2}$.
 (D) Probability that number of diagonals of chosen polygon is an odd number is $\frac{5}{8}$.

Ans. B, C

Sol. Pentagon and square has length of all diagonal equal. Moreover for $n = 5, 6, 9, 10$ no. of diagonals is odd.

There (B) & (C)

Q37. Let m be the median of the data set ${}^{10}C_0, {}^{10}C_1, {}^{10}C_2, \dots, {}^{10}C_{10}$. If

$x_i \in R; i = 1, 2, 3, \dots, n, \sum_{r=1}^n x_i = 80$ and $\sum_{r=1}^n x_i^2 = 400$ then n iCANNOT be equal to :

- (A) $m - 30$ (B) $m - 33$ (C) $m - 27$ (D) $m - 35$

Ans. A, B, D

Sol. -

Q38. a, b, c are three distinct complex numbers such that $(a - b)^5 + (b - c)^5 + (c - a)^5 = 0$, then :

- (A) $\arg\left(\frac{a-b}{b-c}\right) = \frac{\pi}{6}$ (B) $\arg\left(\frac{b-c}{c-a}\right) = \frac{2\pi}{3}$ (C) $|a-b| = |b-c|$ (D) None of these

Ans. B, C

Sol. $z_1 = a - b$

$$z_2 = b - c$$

$$z_3 = c - a$$

$$z_1 + z_2 + z_3 = 0$$

$$z_1^5 + z_2^5 + z_3^5 = 0$$

$$\Rightarrow z_2 = z_1 \omega \text{ \& } z_3 = z_1 \omega^2$$

$$\text{or } z_2 = z_1 \omega^2 \text{ \& } z_3 = z_1 \omega.$$

Q39. For all real number x , let $f(x) = \frac{1}{2x-i} (i = \sqrt{-1})$. If there exists real numbers $a_1, a_2, a_3, a_4, a_5, a_6$ for which $f(a_1), f(a_2), f(a_3), f(a_4), f(a_5), f(a_6)$ form a regular hexagon on the complex plane, then which of the following is/are true :

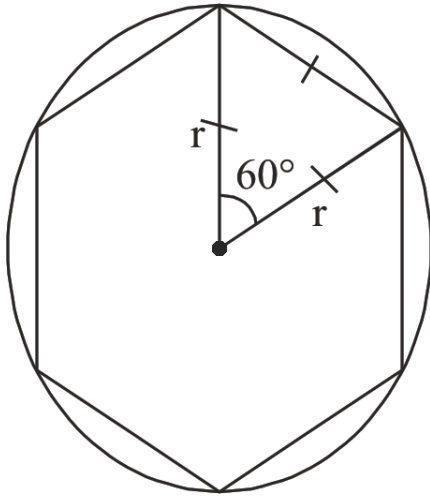
- (A) Area of hexagon is $6\sqrt{3}$ sq. units (B) Area of hexagon is $\frac{3\sqrt{3}}{8}$ sq. units

- (C) Centre of circle circumscribing hexagon is $\left(0, \frac{1}{2}\right)$

- (D) Centre of circle circumscribing hexagon is $\left(0, -\frac{1}{2}\right)$

Ans. B, C

Sol.



$$f(x) = \frac{1}{2x-i} = \frac{2x+i}{4x^2+1}$$

In complex plane, $f(a_1), f(a_2), f(a_3), f(a_4), f(a_5), f(a_6)$ all lies on a curve whose parametric coordinates is given by

$$x = \frac{2t}{4t^2+1}, y = \frac{1}{4t^2+1} \Rightarrow x^2 + y^2 = y$$

$\therefore$ They lies on a circle of radius $1/2$ and centre $(0, 1/2)$.

$\Rightarrow$ Area of hexagon

$$= 6 \left(\frac{\sqrt{3}}{4} r^2 \right)$$

$$= \frac{3\sqrt{3}}{8} \text{ sq. units}$$

Q40. There is a group of 6 persons. They play a game in which each has to select a number from 1 to 4.

Let A_n is event that n persons have selection of same number, then :

$$(A) P(A_5) = \frac{18}{4^6} \quad (B) P(A_5) = \frac{18}{4^5} \quad (C) P(A_6) = \frac{1}{4^5} \quad (D) P\left(\frac{A_5}{A_6}\right) = \frac{1}{2}$$

Ans. B, C

Sol. $P(5 \text{ persons has same selection}) = {}^6C_5 \times {}^4C_1 \times \left(\frac{1}{4}\right)^5 \left(\frac{3}{4}\right)^1$

$$P(6 \text{ persons has same selection}) = {}^4C_1 \times \left(\frac{1}{4}\right)^6$$

$$= \frac{1}{4^5}$$

Also $P\left(\frac{A_5}{A_6}\right) = 0$

Q41. I flip 4 fair coins and eat all of the coins that land on tails. I flip all the uneaten coins again and then eat all the coins that land on tails. The probability that I have eaten at least 3 coins is $\frac{p}{q}$, (where p and q are coprime), then :

$$(A) p = 189 \quad (B) p = 11 \quad (C) q = 16 \quad (D) q = 256$$

Ans. A, D

Sol. Probability is independent, so the probability that any given coin is eaten is $1 - \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{3}{4}$.

The probability of eating 3 or 4 coins is $\binom{4}{3} \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right) + \binom{4}{4} \left(\frac{3}{4}\right)^4 = \frac{108}{256} + \frac{81}{256} = \frac{189}{256}$.

Q42. If $|z| = 1$ and let $\omega = \frac{(1-z)^2}{1-z^2}$, then locus of ω is equivalent to

- (A) $|z - 2 - 4i| = |z - 2 + 4i|$ (B) $|z - 3 + 4i| = |z + 3 + 4i|$ (C) $|z - 2| = |z + 2|$
 (D) $||z - i| - |z + i|| = 2$

Ans. B, C

Sol.
$$\begin{aligned}\omega &= \frac{(1-z)^2}{1-z^2} \\ &= \frac{1-z}{1+z} \\ &= \frac{1-z}{1+z} \cdot \frac{\bar{z}-z}{\bar{z}+z} \quad (\text{As } |z| = 1 \Rightarrow z\bar{z} = 1) \\ \text{Sol. Given } &= \frac{\bar{z}-1}{\bar{z}+1} \\ &= -\left(\frac{1-z}{1+z}\right) \\ &= -\bar{\omega}\end{aligned}$$

$\therefore \omega + \bar{\omega} = 0 \Rightarrow \omega$ is purely imaginary. Hence ω lies on y-axis.

- (A) z lies on perpendicular bisector of $(2, 4)$ and $(2, -4)$ which is x-axis.
 (B) z lies on perpendicular bisector of $(3, -4)$ and $(-3, -4)$ which is y-axis.
 (C) z lies on perpendicular bisector of $(-2, 0)$ and $(2, 0)$ which is y-axis.
 (D) z lies on either of the rays emanating from $(0, 1)$ and $(0, -1)$ towards $+\infty$ and $-\infty$ respectively along y axis, not complete y-axis.

Q43. Let $A(Z_1)$, $B(Z_2)$ and $C(Z_3)$ be the vertices of an equilateral triangle ABC in complex plan then :

- (A) $\frac{1}{Z_1 - Z_2} + \frac{1}{Z_2 - Z_3} + \frac{1}{Z_3 - Z_1} = 0$ (B) $Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_3Z_1$
 (C) $Z_1^3 + Z_2^3 + Z_3^3 = 3Z_1Z_2Z_3$ (D) $\frac{Z_1 - Z_2}{Z_3 - Z_2} = \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$

Ans. A, B, C, D

Sol. $AB = BC = CA$
 $|Z_1 - Z_2| = |Z_2 - Z_3| = |Z_3 - Z_1| = a$
 $\Rightarrow \frac{1}{Z_1 - Z_2} + \frac{1}{Z_2 - Z_3} + \frac{1}{Z_3 - Z_1} = 0$
 $\Rightarrow Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_3Z_1$
 $\angle ABC = \frac{\pi}{3}$
 $\Rightarrow \arg\left(\frac{Z_1 - Z_2}{Z_3 - Z_2}\right)$
 $= \pm \frac{\pi}{3}$

Q44. If z is a complex number satisfying $z^2 + 2z + 4 = 0$ and also $S = \sum_{r=1}^{30} \left(\left(\frac{z}{2} \right)^r + \left(\frac{2}{z} \right)^r \right)^2$, then which of the following is/are true

- (A) $|z| = 1$ (B) $S > 50$ (C) $S < 50$ (D) $\arg(z)$ can be $\frac{2\pi}{3}$

Ans. B, D

Sol. Sol. $z^2 + 2z + 4 = 0$
 roots are 2ω & $2\omega^2$
 so $\frac{z}{2} = \omega$ or $\frac{z}{2} = \omega^2$
 $\left(\frac{z}{2} + \frac{z}{2} \right)^2 = (\omega + \omega^2)^2 = 1$
 $\left(\left(\frac{z}{2} \right)^2 + \left(\frac{2}{z} \right)^2 \right) = (\omega^2 + \omega)^2$
 $= 1$
 $\left(\left(\frac{z}{2} \right)^3 + \left(\frac{2}{z} \right)^3 \right)^2 = (1 + 1)^2$
 $= 4$

Q45. A complex number z is rotated in anticlockwise direction by an angle α and we get z' and if the same complex number z is rotated by an angle α in clockwise direction we get z'' , then :

- (A) z', z, z'' are in G.P. (B) z', z, z'' are in H. P. (C) $z' + z'' = 2z \cos \alpha$
 (D) $z'^2 + z''^2 = 2z^2 \cos 2\alpha$

Ans. A, C, D

Sol. $z' = ze^{i\alpha}$ and $z'' = ze^{-i\alpha}$
 $\therefore z', z'' = z^2$
 z', z, z'' are in G.P.
 Also $\left(\frac{z'}{z} \right)^2 + \left(\frac{z''}{z} \right)^2 = 2 \cos 2\alpha$
 $\Rightarrow z'^2 + z''^2 = 2z^2 \cos 2\alpha = 2z^2 (2 \cos^2 \alpha - 1)$
 $\Rightarrow z'^2 + z''^2 + 2z^2 = 4z^2 \cos^2 \alpha$
 $\Rightarrow z'^2 + z''^2 = 4z^2 \cos^2 \alpha$
 $\Rightarrow (z' + z'')^2 = 4z^2 \cos^2 \alpha$
 $\Rightarrow z' + z'' = \pm 2z \cos \alpha$

Q46. Ram tosses $n + 1$ fair coins and shyam tosses n fair coins ($n \in N$). If $P(n)$ is the probability that Ram gets more heads than Shyam, then :

- (A) $P(n)$ is independent of n (B) $P(n)$ is dependent on n (C) $P(10) = 1/2$
 (D) $P(10) > P(5)$

Ans. A, C

Sol. -

- Q47. Each of 2010 boxes in a line contains one red marble, and for $1 \leq k \leq 2010$, the box at the k^{th} position also contains k white marbles. A child begins at the first box and successively draws a single marble at random from each box in order. He stops when he first draws a red marble. Let $P(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $P(n) < \frac{1}{2010}$, is :

(A) 44 (B) 45 (C) 46 (D) 47

Ans. B, C, D

Sol. **1R 1R 1R 1R 1R 1R → red marbles in the box.**

$B_1 B_2 B_3 B_k B_{2009} B_{2010} \rightarrow$ boxes

$1W 2W 3W kW 200W 2010W \rightarrow$ white marbles in the box.

Now $P(n)$ = probability that child stops after drawing exactly n marbles.

i.e. at the n^{th} position red marble must be drawn.

$$\therefore P(n) = \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{3}{4}\right) \dots \left(\frac{n-2}{n-1}\right) \left(\frac{n-1}{n}\right) \left(\frac{1}{n+1}\right)$$

$$= \frac{1}{n(n+1)}$$

$$\therefore \frac{1}{n(n+1)} < \frac{1}{2010} \Rightarrow \frac{2}{n(n+1)} < \frac{1}{1005}$$

$$\Rightarrow \frac{n(n+1)}{2} > 1005$$

$$\Rightarrow n \geq 45 \Rightarrow \text{B, C, D}$$

- Q48. A box contains 3 coins B_1, B_2 and B_3 . Probability of getting head on these coins are $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{8}$ respectively.

If one of the coins is selected at random and tossed it m times and got exactly 10 heads ($m \geq 10, m \in N$) and x, y, z denote the probabilities that it was coin B_1, B_2, B_3 respectively, then :

(A) $x = \frac{4^{10}}{4^{10} + 2^{10} + 1}$ if $m = 10$ (B) $z = \frac{1}{4^{10} + 2^{10} + 1}$ if $m = 10$ (C) $\lim_{m \rightarrow \infty} x = 0$

(D) $\lim_{m \rightarrow \infty} z = 1$

Ans. A, B, C, D

Sol. Sol.

$$x = P\left(\frac{B_1}{H}\right)$$

$$= \frac{\frac{1}{3} {}^mC_{10} \left(\frac{1}{2}\right)^m}{\frac{1}{3} {}^mC_{10} \left(\frac{1}{2}\right)^m + \frac{1}{3} {}^mC_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{m-10} + \frac{1}{3} {}^mC_{10} \left(\frac{1}{8}\right)^{10} \left(\frac{7}{8}\right)^{m-10}}$$

$$= \frac{4^m}{4^m + 2^m \cdot 3^{m-10} + 7^{m-10}}$$

$$z = P\left(\frac{B_3}{H}\right)$$

$$= \frac{\frac{1}{3} mC_{10} \left(\frac{1}{8}\right)^{10} \left(\frac{7}{8}\right)^{m-10}}{\frac{1}{3} mC_{10} \left(\frac{1}{2}\right)^m + \frac{1}{3} mC_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{m-10} + \frac{1}{3} mC_{10} \left(\frac{1}{8}\right)^{10} \left(\frac{7}{8}\right)^{m-10}}$$

$$= \frac{7^{m-10}}{4^m + 2^m \cdot 3^{m-10} + 7^{m-10}}$$

Now, verify the options.

अब विकल्प को जांचिए

Q49. A slip of paper is given to a person 'A' who marks it with either a plus or a minus sign. The probability of his writing plus is $\frac{1}{3}$. 'A' passes the slip to 'B' who may either leave it as it is or change the sign before passing it to 'C'. Similarly 'C' passes it on to 'D' and 'D' passes it to a referee who sees the plus sign on the slip. If 'B', 'C', 'D' change the sign with probability $\frac{2}{3}$. The probability that A originally wrote the plus sign is :

- (A) less than $\frac{1}{3}$ (B) greater than $\frac{9}{13}$ (C) less than $\frac{13}{25}$ (D) lies in interval $\left(\frac{1}{4}, \frac{1}{2}\right)$

Ans. A, C, D

Sol. Sol. $P(A \text{ wrote } + / \text{ Referee sees } +) = \frac{P(\text{Referee sees } + / A \text{ wrote } +)P(A \text{ wrote } +)}{P(\text{Referee sees } + / A \text{ wrote } +)P(A \text{ wrote } +)}$

$$= \frac{\frac{1}{3} \times \left(\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} \times 3\right)}{\frac{1}{3} \times \left(\frac{1}{27} + \frac{12}{27}\right) + \frac{2}{3} \times \left(\frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \times 3\right)}$$

$$= \frac{\frac{1}{3} \left(\frac{13}{27}\right)}{\frac{1}{3} \left(\frac{13}{27}\right) + \frac{2}{3} \left(\frac{14}{27}\right)}$$

$$= \frac{13}{41}$$

$P(A+ \text{ अंकित करता है / रेफरी + चिन्ह पाता है})$

$$= \frac{P(\text{रेफरी + चिन्ह पाता है} / A, + \text{ अंकित करता है}) P(A, + \text{ अंकित करता है})}{P(\text{रेफरी + चिन्ह पाता है} / A, + \text{ अंकित करता है}) P(A, + \text{ अंकित करता है})}$$

$$= \frac{\frac{1}{3} \times \left(\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} \times 3\right)}{\frac{1}{3} \times \left(\frac{1}{27} + \frac{12}{27}\right) + \frac{2}{3} \times \left(\frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \times 3\right)}$$

$$= \frac{\frac{1}{3} \left(\frac{13}{27}\right)}{\frac{1}{3} \left(\frac{13}{27}\right) + \frac{2}{3} \left(\frac{14}{27}\right)}$$

$$= \frac{13}{41}$$

Paragraph

Consider the equation $x^3 + x^2 - x + 2 = 0$. Let z_1 & z_2 be the imaginary roots & z_3 be the real root of the given equation. Let $2z_1, 2z_2$ & z_3 forms alternate vertices of a regular hexagon & remaining three vertices be α, β, γ . On the basis of above information, answer the following questions :

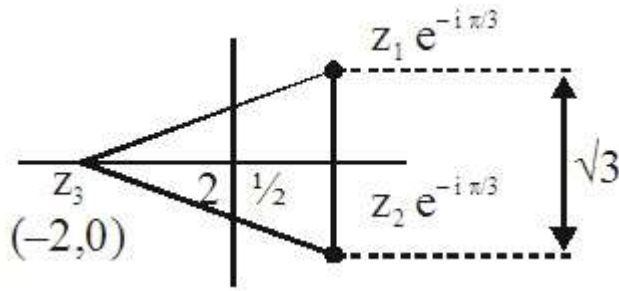
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Q50. The area of the triangle formed by z_1, z_2 & z_3 be -

- (A) $\frac{5}{\sqrt{3}}$ (B) $\frac{5\sqrt{3}}{2}$ (C) $\frac{5\sqrt{3}}{4}$ (D) $5\sqrt{3}$

Ans. C

Sol.



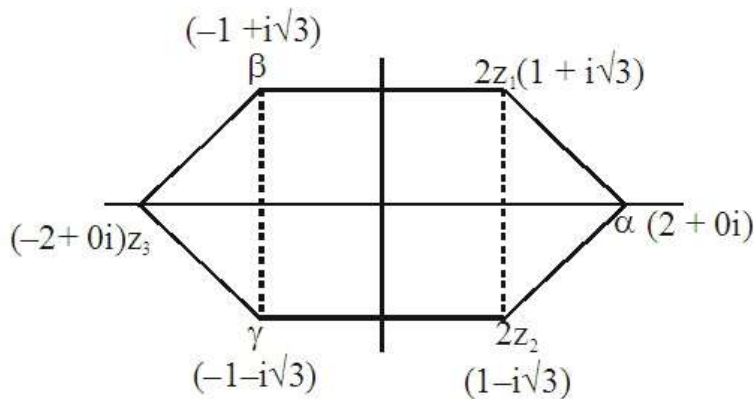
$$A = \frac{1}{2} \times \frac{5}{2} \times \sqrt{3} = \frac{5\sqrt{3}}{4}$$

Q51. If α, β, γ are the roots of equation $x^3 + px^2 + qx + r = 0$, then $(q + r - p)$ is-

- (A) -4 (B) -8 (C) -12 (D) -16

Ans. B

Sol.



Equation whose roots are α, β, γ is

$$(x - 2)(x^2 + 2x + 4) = 0,$$

$$\Rightarrow x^3 - 8 = 0$$

$$\Rightarrow q + r - p = -8$$

Paragraph

$$\text{Let } P = \sum_{r=1}^{50} \frac{{}^{50+r}C_r (2r-1)}{{}^{50}C_r (50+r)}, Q = \sum_{r=0}^{50} ({}^{50}C_r)^2, R = \sum_{r=0}^{100} (-1)^r ({}^{100}C_r)^2 :$$

Q52. The value of $P - Q$ is equal to :

- (A) 1 (B) -1 (C) 2^{50} (D) 2^{100}

Ans. B

Sol. -

Q53. The value of $P - R$ is equal to :

- (A) 1 (B) -1 (C) 2^{50} (D) 2^{100}

Ans. B

Sol. -

Q54. The value of Q + R is equal to :

- (A) $2P + 1$ (B) $2P - 1$ (C) $2P + 2$ (D) $2P - 2$

Ans. C

Sol. General term of the series is

$$T(r) = \sum_{r=1}^{50} \frac{{}^{50+r}C_r (2r-1)}{{}^{50}C_r (50+r)} = \frac{{}^{50+r}C_r}{50C_r} \left(1 - \frac{50-r+1}{50+r} \right) = \frac{{}^{50+r}C_r}{50C_r} - \frac{{}^{50+r}C_r}{50C_r} \left(\frac{50-r+1}{50+r} \right)$$

$$\text{Now, } = \frac{{}^{50+r}C_r}{50C_r} \left(\frac{50-r+1}{50+r} \right) \\ = \frac{(50-r+1)(50+r)!r!(50-r)!}{r!50!(50+r)50!} = \frac{{}^{50+r-1}C_{r-1}}{50C_{r-1}}$$

$$T(r) = \frac{{}^{50+r}C_r}{50C_r} - \frac{{}^{50+r-1}C_{r-1}}{50C_{r-1}} = V(r) - V(r-1) \text{ where } V(r) = \frac{{}^{50+r}C_r}{50C_r}$$

Now, sum of the given series

$$P = \sum_{r=1}^{50} T(r) = V(50) - V(0) = \frac{{}^{100}C_{50}}{50C_{50}} - \frac{{}^{50}C_0}{50C_0} = {}^{100}C_{50} - 1$$

$$\text{Also } C_0^2 - C_1^2 + C_2^2 + \dots + (-1)^n C_n^2 = \begin{cases} 0, & \text{if } n \text{ is odd} \\ (-1)^n C_{n/2}, & \text{if } n \text{ is even} \end{cases}$$

$$\sum (-1)^r ({}^{100}C_r)^2 = (-1)^{100} ({}^{100}C_{50}) = {}^{100}C_{50}$$

$$P - R = -1$$

$$Q + R = 2^{100} C_{50} = 2P + 2$$

Paragraph

If z_0 is complex number and c is a real number, then $\bar{z}_0 z + z_0 \bar{z} + c = 0$ represents a straight line. Its complex slope is $-\frac{z_0}{\bar{z}_0}$. The complex slope of a line passing through two points represented by complex numbers z_1 and z_2 is defined by $m = \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1}$.

Now consider two lines

$$a\bar{z} + \bar{a}z + ib = 0 \quad \dots (1) \quad p\bar{z} + \bar{p}z + q = 0 \quad \dots (2)$$

where a, b and p, q are complex constants and let their complex slopes be denoted by m_1 and m_2 respectively.

Q55. α lies in the interval :

- (A) $[2, 6]$ (B) $[1, 7/2]$ (C) $[2, 5]$ (D) $[-4, -2]$

Ans. C

Sol. -

Q56. Maximum value of $|z - i| =$:

- (A) $\frac{7}{2}$ (B) $\frac{4}{9}$ (C) $\frac{5}{9}$ (D) $\frac{9}{4}$

Ans. D

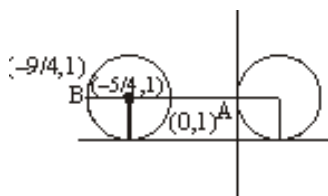
Sol. -

Q57. The value of $\arg(z)$ for which $|z - i|$ is maximum is :

- (A) $\pi - \tan^{-1} (4/9)$ (B) $\pi - \tan^{-1} (9/4)$ (C) $\pi - \tan^{-1} (1/9)$ (D) $\pi - \tan^{-1} (4/5)$

Ans. B

Sol.



$$\alpha^2 - 7\alpha + 11 \leq 1$$

for some part to be in 2nd quadrant $\Rightarrow a \in [2, 5]$

$$\therefore (\alpha^2 - 7\alpha + 11)_{\min} = -5/4$$

$$AB = \frac{9}{4} \text{ and } \arg z = \pi - \tan^{-1} \left(\frac{9}{4} \right)$$

Paragraph

In an objective paper, there are two sections of 10 questions each. For 'section 1', each question has 5 options and only one option is correct and 'section 2' has 4 options with multiple answers and marks for a question in this section is awarded only if he ticks all correct answers. Marks for each question in 'section 1' is 1 and in 'section 2' is 3. (There is no negative marking.)

Q58. If a candidate in total attempts 4 questions all by guessing, then the probability of scoring 10 marks is :

- (A) $\frac{1}{15} \left(\frac{1}{15} \right)^3$ (B) $\frac{4}{5} \left(\frac{1}{15} \right)^3$ (C) $\frac{1}{5} \left(\frac{14}{15} \right)^3$ (D) None of these

Ans. D

Sol. Scoring 10 marks from four questions can be done in $3 + 3 + 3 + 1 = 10$ ways so as to answer 3 questions from section 2 and 1 question from section 1 correctly. Hence the required probability is

$$\frac{{}^{10}C_3 {}^{10}C_1}{20C_4} \cdot \frac{1}{5} \left(\frac{1}{15} \right)^3$$

Paragraph

P is a power set of a given set S containing n elements and $A, B, C \in P$ (may or may not be distinct).

Q59. If A, B and C are disjoint sets find probability that $A \cup B \cup C = S$:

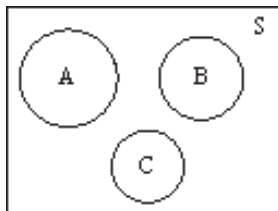
- (A) $\left(\frac{4}{7} \right)^{n-1}$ (B) $\left(\frac{3}{4} \right)^n$ (C) $\left(\frac{3}{7} \right)^n$ (D) $\left(\frac{5}{7} \right)^n$

Ans. B

Sol. If A, B and C are disjoint sets then number of ways to place n elements is 4^n

If A, B and C are disjoint sets and $A \cup B \cup C = S$ then number of ways to place n elements is

3^n .

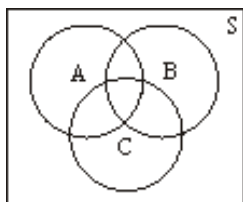


Q60. If $A \cap B \cap C$ contains exactly 1 element, find probability that set B also contain exactly 1 element :

- (A) $\left(\frac{4}{7}\right)^{n-1}$ (B) $\left(\frac{3}{4}\right)^n$ (C) $\left(\frac{3}{7}\right)^n$ (D) $\left(\frac{5}{7}\right)^n$

Ans. A

Sol. If $A \cap B \cap C$ contain exactly 1 element, then number of ways to place n elements is $n \times 7^{n-1}$ while if B also contain exactly 1 element, then number of ways to place n elements $n \times 4^{n-1}$.

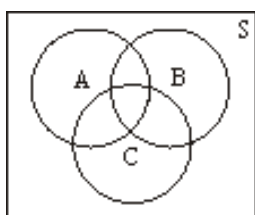


Q61. If $A \cap B \cap C$ contain no element, find probability that $A = B$:

- (A) $\left(\frac{4}{7}\right)^{n-1}$ (B) $\left(\frac{3}{4}\right)^n$ (C) $\left(\frac{3}{7}\right)^n$ (D) $\left(\frac{5}{7}\right)^n$

Ans. C

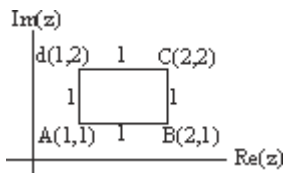
Sol. If $A \cap B \cap C$ contain no element, then number of ways to place n elements is 7^n while if $A \cap B \cap C$ contain no element and $A = B$ then number of ways to place n elements is 3^n .



Q62. The area bounded by the rays $\arg(z - (1 + i)) = 0$, $\arg(z - (2 + i)) = \frac{\pi}{2}$, $\arg(z - (2 + 2i)) = \pi$ and $\arg(z - (1 + 2i)) = -\frac{\pi}{2}$ is λ , then the value of $\frac{\lambda}{2}$ is _____.

Ans. 0.50

Sol.



Im(z)

d(1,2) 1 C(2,2)

1 1

A(1,1) 1 B(2,1)

Re(z)

area = $|x| = 1$

$\lambda = 0.5$

AB is represented by $\arg(z - (1 + i)) = 0$, BC

is represented by $\arg(z - (2 + i)) = \frac{\pi}{2}$,

CD is represented by $\arg(z - (2 + 2i)) = \pi$, DA

is represented by $\arg(z - (1 + 2i)) = -\frac{\pi}{2}$

Q63. If z_1 and z_2 are two complex number such that $z_1 + z_2 = z_1^2 + z_2^2 = \frac{2i}{\sqrt{3}}$ and $|\operatorname{Re}(z_1)| = \sin\left(\frac{\pi}{k}\right)$ where $i = \sqrt{-1}$, then the value of $\frac{1}{k}$ is equal to _____.

Ans. 0.25

Sol. $z_1 + z_2 = z_1^2 + z_2^2 = \frac{2i}{\sqrt{3}}$ _____ (1)

$$\Rightarrow (z_1 + z_2)^2 - 2z_1z_2 = \frac{2i}{\sqrt{3}}$$

$$\Rightarrow z_1z_2 = -\frac{2}{3} - \frac{i}{\sqrt{3}} \text{ _____ (2)}$$

$$\text{Let } z_1 = x + iy \text{ and } z_2 = -x + i\beta \left\{ \because z_1 + z_2 = \frac{2i}{\sqrt{3}} \right\}$$

From eq. (2)

$$(-x^2 - y\beta) = -\frac{2}{3}, x(\beta - y) = -\frac{1}{\sqrt{3}}$$

From eq. (1)

$$\beta + y = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \beta = \frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{3}x}$$

$$= \frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{3}x}$$

Putting the value of β and y

$$12x^4 - 4x^2 - 1 = 0 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$\operatorname{Re}(z_1) = \sin\left(\frac{\pi}{4}\right) \Rightarrow k = 4$$

Q64. If the line $2x + y = k$ passes through the point which divides the line segment joining the points $(1, 1)$ and $(2, 4)$ in the ratio $3 : 2$, then k equal 2λ . If $|z_1| = 2$, $|z_2| = \lambda$, $|z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 3\lambda$, then value of $|8z_2z_3 + 27z_3z_1 + 64z_1z_2|$ is equal to (where z_1, z_2, z_3 are complex numbers) :

Ans. 216

Sol.
$$\left| \frac{z_1 z_2 z_3}{z_1 z_2 z_3} \left(\frac{2|z_1|^2}{z_1} + \frac{3|z_2|^2}{z_2} + \frac{4|z_3|^2}{z_3} \right) \right|$$

$$|z_1 z_2 z_3| |2\overline{z_1} + 3\overline{z_2} + 4\overline{z_3}|$$

$$2 \times 3 \times 4 \times 9 = 216$$

Q65. In an n -by m grid, 1 row and 1 column are colored blue, the rest of the cells are white. If precisely $\frac{1}{2010}$ of the cells in the grid are blue, how many values are possible for the ordered pair (n, m) ?

Ans. 96

Sol. The number of blue cells is $n + m - 1$; the number of total cells is nm . So $2010(m + n - 1) = nm$, or $nm - 2010(m + n) = -2010$. Thus each of $n - 2010$ and $m - 2010$ must be one of the positive factors of 2010×2009 ; for each positive factor, there is one ordered pair. Since $2010 \cdot 2009 = 2 \cdot 3 \cdot 5 \cdot 7^2 \cdot 41 \cdot 67$, there are $2 \cdot 2 \cdot 2 \cdot 3 \cdot 2 \cdot 2 = 2^5 \cdot 3 = 96$ solutions.

Q66. Four cards are drawn from a standard deck (52 cards) with suits indistinguishable (for example, A of hearts is the same as A of spades). How many distinct hands can one obtain?

Ans. 1820

Sol. We proceed by casework.

Case-1 All cards have the same face value. There are $\binom{13}{1}$ ways to choose the face values.

Case-2 Some cards have face value A; some have face value B. There are $\binom{13}{2}$ ways to choose A and B.

One can have the combination ABBB, AABB, AAAB, so there are $3 \binom{13}{2}$ distinct ways for this case.

Case-3 Some cards have value A, some B and C. There are $\binom{13}{3}$ ways to choose the A, B, C. One can

have the combination ABCC, ABBC, and AABC. There are $3 \binom{13}{3}$ distinct ways for this case.

Case-4 The cards are distinct: ABCD. There are $\binom{13}{4}$ ways to do this. Since these cases are mutually

exclusive, we have $\binom{13}{1} + 3 \binom{13}{2} + 3 \binom{13}{3} + \binom{13}{4} = 1820$ distinct hands.

Q67. If $\sum_{r=1}^n |r \cdot |r - 1| (r^2 + r - 1)| = |9|10 - 1|$, then value of (n^2) is equal to _____.

Ans. 81

Sol.
$$\sum_{r=1}^n |r| |r - 1| (r^2 + r - 1)$$

$$\sum_{r=1}^n (|r| |r + 1| - |r| |r - 1|)$$

 now do yourself

$$\sum_{r=1}^n |r|r - 1| (r(r+1) - 1)$$

$$\sum_{r=1}^n (|r|r + 1| - |r|r - 1|)$$

now do yourself

- Q68. A die is rolled four times. If the probability that product of first 3 outcomes is equal to fourth outcome is p , then $\left(\frac{1}{\sqrt{p}}\right)$ is equal to _____.

Ans. 7.2

Sol.
$$p = \left(\frac{1}{6}\right)^3 \frac{1}{6} + 3\left(\frac{1}{6}\right)^3 \frac{1}{6} + 3\left(\frac{1}{6}\right)^3 \frac{1}{6} + 6\left(\frac{1}{6}\right)^3 \frac{1}{6} + 3\left(\frac{1}{6}\right)^3 \frac{1}{6} + 9\left(\frac{1}{6}\right)^3 \frac{1}{6}$$

$$= \frac{25}{6^4}$$

- Q69. The marks obtained by 40 students are grouped in a frequency table in class intervals of 10 marks each. The mean and variance obtained from this distribution are found to be 40 and 49 respectively. It was later discovered that the two observations belonging to class interval (21-30) were included in the the class interval (31-40) by mistake. Then the mean after correcting the error is equal to _____.

Ans. 39.5

Sol. -

- Q70. Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression with $x_3 = \frac{3}{4}$ and $x_1 = 3$. A new data set is constructed by replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ denotes the mean of new data then $\lambda = \max\{a \in \mathbb{Z} : a \leq \bar{x}\}$ is equal to _____.

Ans. 142

Sol. -

- Q71. Let $P(z) = z^3 + az^2 + bz + c$, where a, b, c are real. There exist a complex number w such that the three roots of $P(z)$ are $w + 3i$, $w + 9i$ and $2w - 4$, where $i^2 = -1$. If the value of $|a + b + c|$ is written as $2^{n_1} \cdot 3^{n_2} \cdot 5^{n_3} \cdot 7^{n_4}$, then value of $n_1 + n_2 + n_3 + n_4$ is equal to _____.

Ans. 4

Sol. . a, b, c are real numbers, so two roots are complex conjugate of each other and one root is real.

Let $w + 3i$ is real. Let $w + 3i = a$

$$\Rightarrow w = a - 3i$$

so, roots are $a, a + 6i, 2(a - 3i) - 4$

i.e. $a, a + 6i, (2a - 4) - 6i$

complex conjugate of each other

$$\text{so } a = 2a - 4 \Rightarrow a = 4.$$

. roots are $4, 4 + 6i, 4 - 6i$ (other cases are not possible)

$$\text{. equation is } (z - 4)(z - (4 + 6i))(z - (4 - 6i)) = 0$$

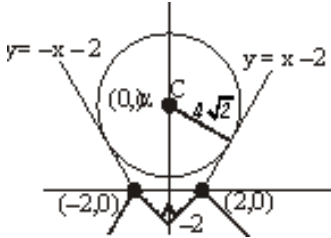
$$\Rightarrow z^3 - 12z^2 + 84z - 208 = 0$$

$$\therefore |a + b + c| = |-12 + 84 - 208| = 136.$$

Q72. Consider curves $C_1 : |\arg(z - 2)| = \frac{\pi}{4}$; $C_2 : |\arg(z + 2)| = \frac{3\pi}{4}$ and $C_3 : |z - \alpha i| = 4\sqrt{2}$, $\alpha \in R$ on the complex plane such that C_3 touches both C_1 and C_2 , then find the sum of absolute value(s) of α _____.

Ans. 12

Sol.



$$\alpha + 2 = 8$$

$$\alpha = 6$$

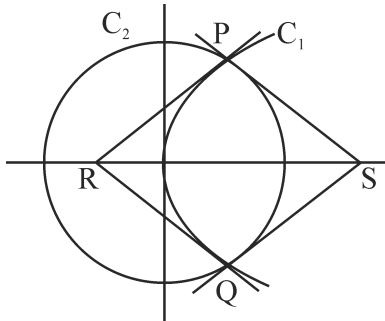
similarly इसी प्रकार $\alpha = -6$

$\therefore$ sum योगफल = 12

Q73. Consider the curves $C_1 : |z - 2| = 2 + \operatorname{Re}(z)$ and $C_2 : |z| = 3$ (where $z = x + iy$, $x, y \in R$ and $i = \sqrt{-1}$). They intersect at P and Q in the first and fourth quadrants respectively. Tangents to C_1 at P and Q intersect the x-axis at R and tangents to C_2 at P and Q intersect the x-axis at S. If area of $\triangle PRS$ is $\lambda\sqrt{2}$ sq. units, then (λ^2) is _____.

Ans. 100

Sol.



Put $z = x + iy$

$$C_1 : y^2 = 8x$$

$$C_2 : x^2 + y^2 = 9$$

On solving C_1 & C_2 $P(1, 2\sqrt{2})$

$$\text{Tangent to } C_1 \text{ at } P : y(2\sqrt{2}) = 8 \frac{(x+1)}{2}$$

$$\text{at } y = 0 : x = -1 \Rightarrow R(-1, 0)$$

$$\text{Tangent to } C_2 \text{ at } P : x(1) + y(2\sqrt{2}) = 9$$

$$\text{at } y = 0 : x = 9 \Rightarrow S(9, 0)$$

$$\therefore \text{Area } \triangle PRS : \frac{1}{2} \times (9 - (-1)) \times 2\sqrt{2} = 10\sqrt{2}$$

$$\therefore \lambda = 10 \Rightarrow \lambda^2 = 100$$

Q74. In a tournament, team X, plays with each of the 6 other teams once. For each match the probabilities of a win, draw and loss are equal the probability that the team X, finishes with

more wins than losses is m/n where m, n are coprime. Let $A = [a_{ij}]$ be a matrix of order 2 where $a_{ij} \in \{-1, 0, 1\}$ and $\text{adj. } A = -A$. If $\det(A) = -1$ and the number of such matrices are p then find the value of $3m - n - 3p$ is _____.

Ans. 3

Sol. $\frac{98}{243}$

$$\text{adj. } A = -A \Rightarrow A \cdot \text{adj. } A = -A^2$$

$$\text{Let } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{So, } A^2 = I$$

$$\Rightarrow \begin{pmatrix} a^2 + bc & (a+d)b \\ (a+d)c & d^2 + bc \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\therefore$ On comparing, we get

$$a^2 + bc = 1, (a+d)b = 0,$$

$$(a+d)c = 0, d^2 + bc = 1$$

Case I: When $(a+d) \neq 0$

$$\Rightarrow b = 0 \text{ and } c = 0, a = 1, d = 1$$

$$\text{or } a = -1, d = -1$$

$$\therefore A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ or } \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

but both rejected as $\det. A = -1$ (given)

$$(i) \text{ If } a = 1, d = -1 \Rightarrow bc = 0$$

For $b = 0$, c can be $-1, 0, 1$

For $b = 1$, c can be 0 only

For $b = -1$, c can be 0 only

$\Rightarrow 5$ matrices

$$(ii) \text{ If } a = -1, d = 1 \Rightarrow bc = 0$$

For $b = 0$, $c = -1, 0, 1$

For $b = 1$, $c = 0$ only

For $b = -1$, $c = 0$ only

$\Rightarrow 5$ matrices

$$(iii) \text{ If } a = 0, d = 0 \Rightarrow bc = 1$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ or } A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

$\Rightarrow 2$ matrices

$$\therefore N = 5 + 5 + 2 = 12$$

$$\Rightarrow (N - 7) = 12 - 7 = 5$$

Q75. The number of ordered pairs (m, n) ; $m, n \in \{1, 2, 3, \dots, 20\}$ such that $3^m + 7^n$ is a multiple of 10, is equal to _____.

Ans. 100

Sol. End digits in power of 3 and 7 repeat in cycle of 4

$$3^m \rightarrow 3^{4k} = 1, 3^{4k+1} = 3, 3^{4k+2} = 9, 3^{4k+3} = 7$$

$$7^n \rightarrow 7^{4k} = 1, 7^{4k+1} = 7, 7^{4k+2} = 9, 7^{4k+3} = 3$$

So the number 1, 2, 3, ..., 20 can be divided into 4 sets

$$4k \text{ type } 4, 8, 12, 16, 20 = A$$

$$4k + 1 \text{ type } 1, 5, 9, 13, 17 = B$$

$$4k + 2 \text{ type } 2, 6, 10, 14, 18 = C$$

$$4k + 3 \text{ type } 3, 7, 11, 15, 19 = D$$

Now $3^m + 7^n$ divisible by 10, if end digit of the number is 0.

Thus if $m \in A$ then $n \in C$, if $m \in B$ then $n \in B$

If $m \in C$ then $n \in A$, if $m \in D$ then $n \in D$

So, desired number of ordered pairs (m, n)

$$= 5 \times 5 + 5 \times 5 + 5 \times 5 + 5 \times 5 = 100$$

Q76. Let n be a number amongst the first hundred natural numbers such that

$(\sin \theta + i \cos \theta)^n = \sin(n\theta) + i \cos(n\theta)$, then find number of possible values of n _____.

Ans. 25

Sol. $\left\{ \cos \left(\frac{\pi}{2} - \theta \right) + i \sin \left(\frac{\pi}{2} - \theta \right) \right\}^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$ (DeMoivre's Theorem
द:मायावर प्रमेय)

$$= \sin(n\theta) + i \cos(n\theta)$$

Hence अतः $n = 4k + 1$

$$\Rightarrow n = 1, 5, 9, \dots, 97$$

$\Rightarrow$ Number of value is मानों की संख्या 25 है।

Q77. If $A = \left\{ z : \left| \frac{z-2}{z+2} \right| = 3, z \in C \right\}$ and $z_1, z_2, z_3, z_4 \in A$ are 4 complex number representing points P, Q, R, S respectively on the complex plane such that $z_1 - z_2 = z_4 - z_3$, then maximum value of area of quadrilateral PQRS is _____.

Ans. 4.5

Sol. Let $z = x + iy$

$$\Rightarrow |x - 2 + iy| = 3|x + iy + 2|$$

$$\Rightarrow (x - 2)^2 + y^2 = 9(x + 2)^2 + 9y^2$$

$$\Rightarrow 8x^2 + 8y^2 + 40x + 32 = 0$$

$$\Rightarrow x^2 + y^2 + 5x + 4 = 0$$

Clearly locus is a circle with centre $\left(-\frac{5}{2}, 0 \right)$ and radius $\sqrt{\frac{24}{4}} = \frac{3}{2}$

since $z_1, z_2, z_3, z_4 \in A$

$\Rightarrow$ all these points must lie on this circle and forms a parallelogram, since $z_1 + z_3 = z_2 + z_4$

Now to have maximum area of parallelogram, it must be a square whose area will be

$$\frac{d^2}{2} = \frac{9}{2}$$



Q78. In each of a set of games it is 2 to 1 in favour of the winner of the previous game. the chance that the player who wins the first game shall win three at least, of the next four is m/n where m, n are

coprime. If $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) = \tan^{-1} \frac{m}{x}$ then the value of $250x^4 + 320x^2 - 1630$ is equal to _____.

Ans. 10

Sol. We have

$$\frac{x + \frac{2}{x} - \left(x - \frac{2}{x}\right)}{1 + \left(x^2 - \frac{4}{x^2}\right)} = \frac{4}{x}$$

$$\Rightarrow x^2 - \frac{4}{x^2} = 0$$

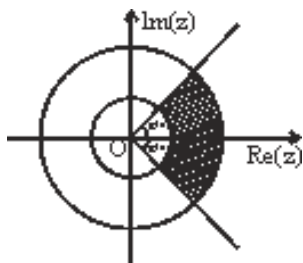
$$\Rightarrow x^2 = 2$$

$$\text{So } (250x^4 + 320x^2 - 1630) = 250 \times 4 + 320 \times 2 - 1630 = 10$$

Q79. Let complex number 'z' satisfy the inequality $2 \leq |z| \leq 4$. A point P is selected in this region at random. The probability that argument of P lies in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is $\frac{1}{K}$, then $K =$ _____.

Ans. 4

Sol.



$$2 \leq |z| \leq 4$$

$$\text{Probability} = \frac{1}{4}$$

Q80. Number of ways in which the letters of word ABBCABBC can be arranged such that the word ABBC does not appear in any word, is a three digit natural number, then find the units digit of this number _____.

Ans. 1

Sol. Total words formed = $\frac{8!}{4! 2! 2!} = 420$

Let ABBC = $\times$ Number of ways in which $\times$ ABBC can be arranged = $\frac{5!}{2!} = 60$ but this includes $\times$ ABBC

and ABBC $\times$. But the word ABBCABBC is counted twice in 60 hence it should be 59 so required number

$$\text{of ways} = 420 - 59 = 361$$

Q81. Find the positive integer just greater than $(1 + 0.0001)^{10000}$

Ans. 3

$$\begin{aligned} \text{Sol. } (1 + .0001)^{10000} &= 1 + 10000(.0001) + \frac{10000(10000-1)}{2!}(.0001)^2 + \dots \\ &= 1 + 10000(.0001) + \frac{10000(9999)}{2}(.0001)^2 + \dots \\ &= 1 + 1 + \frac{10000(9999)}{2}(.0001)^2 + \dots \end{aligned}$$

$$= 2 + 10000 C_2 (.0001)^2 + 10000 C_3 (.0001)^3 + \dots$$

$$\Rightarrow (1 + .0001)^{10000} \in (2, 3)$$

$$\Rightarrow \text{positive integer just greater than}$$

Q82. Four fair dice are thrown simultaneously. If probability that the highest number obtained is 4 is $\frac{25a}{1296}$ then 'a' is equal to _____.

Ans. 7

Sol. 4 must be obtained on atleast one dice

x = number of dice on which 4 is obtained

required probability

$$= P(x = 1) + P(x = 2) + P(x = 3) + P(x = 4)$$

$$= \frac{{}^4C_1 \times 3 \times 3 \times 3}{6^4} + \frac{{}^4C_2 \times 3^2}{6^4} + \frac{{}^4C_3 \times 3}{6^4} + \frac{1}{6^4}$$

$$= \frac{175}{1296}$$

Q83. A box contains 28 identical balls of which 21 are red and 7 are black. The balls are drawn at random from the box one at a time with replacement. If the Probability that red ball is drawn for the 5th time on the 13th draw is $\frac{55 \cdot 3^k}{4^{13}}$, then k is _____.

Ans. 9

Sol. $P(R) = \frac{3}{4}, P(B) = \frac{1}{4}$

$$\text{Required probability} = \frac{12!}{8!4!} \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^8 \cdot \frac{3}{4}$$

Q84. Mr. A writes an article. The article originally is error free. Each day Mr. B introduces one new error into the article. At the end of the day, Mr. A checks the article and has $\frac{2}{3}$ chance of catching each individual error still in the article. After 3 days, the probability that the article is error free can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive integers. Let $\lambda = q - p$. Then find the sum of the digits of λ .

Ans. 7

Sol. $\left(\frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3}\right) \left(\frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3}\right) = \frac{416}{729}$

Q85. A construction company has three different sites located in the city. Analysis of the daily wages paid to the labourers in the three sites are as follows.

	Site 1	Site 2	Site 3
No. of labourers	20	25	40
Average daily wages (Rs)	305	300	340
Standard deviation (Rs)	50	40	45

If the standard deviation of the wages of all the 85 labourers in the city is given by Rs $\sqrt{\frac{p}{q}}$, where $p, q \in \mathbb{N}$, H. C. F (p,q) = 1, then find the value of $\left[\frac{p}{q} - 2362 \right]$ (where $[.]$ denotes the greatest integer function) _____.

Ans. 8

Sol. -

Q86. Let α and β are distinct reals satisfying $\sum_{i=1}^{18} (x_i - \alpha) = 36$ and $\sum_{i=1}^{18} (x_i - \beta)^2 = 90$. If the standard deviation of $x_1, x_2, x_3, \dots, x_{18}$ is 1, then the value of $|\alpha - \beta|$ is equal to _____.

Ans. 4

Sol. -

Q87. If $n = 12m$ ($m \in \mathbb{N}$), ${}^nC_0 - \frac{{}^nC_2}{(2 + \sqrt{3})^2} + \frac{{}^nC_4}{(2 + \sqrt{3})^4} - \frac{{}^nC_6}{(2 + \sqrt{3})^6} + \dots = (-1)^m \left(\frac{p\sqrt{p}}{1 + \sqrt{q}} \right)^n$ then find the value of $p + q$. _____.

Ans. 5

Sol. $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

Also, $(\cos \theta + i \sin \theta)^n = {}^nC_0 \cos^n \theta + {}^nC_1 \cos^{n-1} \theta (i \sin \theta) + {}^nC_2 \cos^{n-2} \theta (i \sin \theta)^2 + \dots$

Equating the real parts, we get

$\cos n\theta = {}^nC_0 \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta + \dots$

or $\frac{\cos n\theta}{\cos^n \theta} = {}^nC_0 - {}^nC_2 \tan^2 \theta + {}^nC_4 \tan^4 \theta - {}^nC_6 \tan^6 \theta + \dots$

put $\theta = 15^\circ$, we get $\tan \theta = 2 - \sqrt{3} = \frac{1}{2 + \sqrt{3}}$ and $\cos n\theta = \cos 12m \frac{\pi}{12} = \cos m\pi = (-1)^m$

and $\cos \theta = \frac{\sqrt{3} + 1}{2\sqrt{2}}$

So, ${}^nC_0 - \frac{{}^nC_2}{(2 + \sqrt{3})^2} + \frac{{}^nC_4}{(2 + \sqrt{3})^4} - \frac{{}^nC_6}{(2 + \sqrt{3})^6} + \dots = (-1)^m \left(\frac{2\sqrt{2}}{1 + \sqrt{3}} \right)^n$

Q88. In a tournament four players are participating. Each player play with every other player. Each player has 50% chance of winning any game and there are no ties. If the probability that at the end of tournament there is neither a winless nor an undefeated player is there is $\frac{a}{b}$, where a and b are relatively prime integers then $|2a - b|$ is equal to _____.

Ans. 2

Sol. A : There is at least one player with all wins

B : There is at least one player with all defeats

E : neither a winless nor undefeated player.

$P(E) = 1 - P(A \cup B)$

$$1 - \left[{}^4C_1 \left(\frac{1}{2} \right)^3 + {}^4C_1 \left(\frac{1}{2} \right)^3 - {}^4C_1 \left(\frac{1}{2} \right)^3 {}^3C_1 \left(\frac{1}{2} \right)^2 \right]$$

$$1 - \left[\frac{4}{8} + \frac{4}{8} - \frac{4.3}{8.4} \right] = 1 - \left(1 - \frac{3}{8} \right) = \frac{3}{8}$$

Q89. Six people play the following game : They have a cube, initially white. Once by one, the players mark an X on a white face of the cube, and roll it like a die. The winner is the first person to roll an X (for example, player 1 wins with probability $\frac{1}{6}$, while if none of players 1 – 5 win, player 6 will place an X on the last white square and win for sure). The probability that the sixth player wins is $\frac{n}{324}$. Find the value of n
_____.

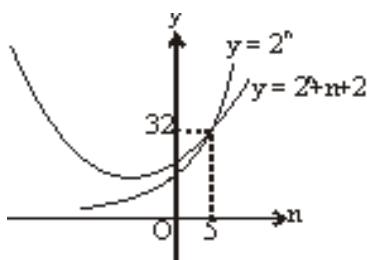
Ans. 5

Sol. The probability of player n winning if no one has already done so is $\frac{n}{6}$. Restated, the probability that player n will NOT win if no one has already done so is $\frac{6-n}{6}$. So the probability that the first 5 players will all lose is $\frac{5}{6} \cdot \frac{4}{6} \cdot \frac{3}{6} \cdot \frac{2}{6} \cdot \frac{1}{6} = \frac{5}{324}$, and this is the only way the sixth player can win.

Q90. N fair coins are flipped once. The probability that at most 2 of the coins show up as heads is $\frac{1}{2}$. Find the value of N _____.

Ans. 5

Sol.



$$P(0 \text{ heads}) = \left(\frac{1}{2} \right)^N$$

$$P(1 \text{ heads}) = \left(\frac{1}{2} \right) \cdot \left(\frac{1}{2} \right)^{N-1} \binom{N}{1} = N \left(\frac{1}{2} \right)^N$$

$$P(2 \text{ heads}) = \left(\frac{1}{2} \right)^2 \cdot \left(\frac{1}{2} \right)^{N-2} \binom{N}{2} = \frac{N(N-1)}{2} \left(\frac{1}{2} \right)^N$$

$$\therefore P(\text{at most 2 heads}) = P(0 \text{ heads}) + P(1 \text{ head}) + P(2 \text{ heads})$$

$$\Rightarrow \frac{1}{2}$$

$$= \left(\frac{1}{2} \right)^N + N \left(\frac{1}{2} \right)^N + \frac{N(N-1)}{2} \left(\frac{1}{2} \right)^N$$

$$\Rightarrow N^2 + N + 2 = 2^N \Rightarrow N = 5$$

$$\begin{aligned}
 P(0 \text{ heads}) + P(1 \text{ head}) + P(2 \text{ heads}) & \Rightarrow \frac{1}{2} \\
 & = \left(\frac{1}{2}\right)^N + N\left(\frac{1}{2}\right)^N + \frac{N(N-1)}{2}\left(\frac{1}{2}\right)^N \\
 \Rightarrow N^2 + N + 2 & = 2^N \Rightarrow N = 5
 \end{aligned}$$

Q91. A person flipped a fair coin five times. In the first three flips, the coins came up heads exactly twice and in the last three flips also, the coin came up heads exactly twice. The probability that the third flip was a heads, is $\frac{p}{q}$ (where p and q are coprime) the find the remainder when p^{3q} is divided by 7.

Ans. 1

Sol. Equally like, mutually exclusive and exhaustive cases to have exactly 2H in first three and exactly 2H in last three, when the coin is thrown 5 times, are as shown

$$\begin{pmatrix} H & H & T & H & H \\ H & T & H & H & T \\ H & T & H & T & H \\ T & H & H & H & T \\ T & H & H & T & H \end{pmatrix}$$

Hence $P(E) = 4/5$

Q92. The number of ways of arranging 11 objects A, B, C, D, E, F, $\alpha, \alpha, \alpha, \beta, \beta$ so that every β lie between two α (not necessarily adjacent) is $K \times 6! \times {}^{11}C_5$, then K is _____

Ans. 3

Sol. There are three major ways $\alpha\alpha\beta\beta\alpha, \alpha\beta\beta\alpha\alpha$ and $\alpha\beta\alpha\beta\alpha$

Each major way has six empty spaces. The number of ways of putting letters at these empty spaces must be non-negative integer function of $x_1 + x_2 + \dots + x_6 = 6 = {}^{6+6-1}C_{6-1} = {}^{11}C_5$

No. of arrangements is $= 3 \times {}^{11}C_5 \times 6! \Rightarrow K = 3$

Q93. A complex number z satisfies the system of equations $|z - 4| = |z - 4 - 3i|, |z - 2 + 3i| = |z - 5i|$, then $2\operatorname{Re}(z) =$ _____.

Ans. 6

Sol. $|z - 4| = |z - 4 - 3i|$

$\Rightarrow z$ lies on $\perp$ bisector $y = \frac{3}{2}$

$|z - 2 + 3i| = |z - 5i|$

$\Rightarrow z$ lies on $\perp$ bisector $\Rightarrow 4y = x + 3$

Q94. If $A(z_1), B(z_2), C(z_3)$ are the vertices of an equilateral triangle ABC. If the value of

$\left| \arg \left(\frac{z_2 + z_3 - 2z_1}{2z_1 - 4z_2 + 2z_3} \right) \right|$ is $\frac{P\pi}{Q}$ (P and Q are in their lowest form), then find the value of $(P + Q)$

_____.

Ans. 5

Sol. $\left| \arg \left(\frac{z_2 + z_3 - 2z_1}{2z_1 - 4z_2 + 2z_3} \right) \right|$
 $= \left| \arg \left(\frac{2 \left(\frac{z_2 + z_3}{2} - z_1 \right)}{4 \left(\frac{z_1 + z_3}{2} - z_2 \right)} \right) \right|$
 $= \left| \arg \left(\frac{\left(\frac{z_2 + z_3}{2} - z_1 \right)}{\left(\frac{z_1 + z_3}{2} - z_2 \right)} \right) \right|$
 $= \frac{2\pi}{3}$



Q95. If $z_1, z_2, z_3, z_4, z_5, z_6$ are the roots of the equation $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$, then the value of $\sum_{i=1}^6 z_i^5 + \sum_{i=1}^6 z_i^{14} - \prod_{i=1}^6 z_i$ is equal to _____.

Ans. 4

Sol. Given equation

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0 \dots (i)$$

$$\therefore \text{Product of roots} = \prod_{i=1}^6 z_i = 1$$

Now, multiply equation (1) by $(z - 1)$, we get

$$z^7 - 1 = 0 \Rightarrow z^7 = 1 \dots (2) \text{ i.e. } 7^{\text{th}} \text{ roots of unity.}$$

$\therefore$ roots of equation (1) are the roots of equation (2) except $z = 1$.

$$\therefore \sum_{i=1}^6 z_i^5 = -1 \text{ and } \sum_{i=1}^6 z_i^{14} = 6$$

$$\therefore \text{required sum} = -1 + 6 - 1 = 4$$

Q96. Consider lines $z(1 - mi) - \bar{z}(1 + mi) = 0$ and $z(1 - ni) - \bar{z}(1 + ni) = 0$. Let $\arg(z')$ be the angle between them such that $\arg(z') \in \left[0, \frac{\pi}{2}\right]$. There is another line $z(1 + i) - \bar{z}(-1 + i) + 2i = 0$ which intersects the previous lines in A and B such that $AB = OA = OB = AB$. Then, $\text{Max}(\arg(z'))$ is $\frac{\pi}{k}$, where k is _____.

Ans. 3

Sol. $\theta = \arg(z')$

$$\cos \theta = \frac{OA^2 + OB^2 - AB^2}{2.OA.OB}$$

$$\cos \theta = \frac{(OA)^2 + (OB)^2 - \frac{1}{4}(OA + OB)^2}{2.OA.OB}$$

$$= \frac{8}{3} \left(\frac{OA}{OB} + \frac{OB}{OA} \right) - \frac{1}{4}$$

$$\text{For } \theta \text{ (Maximum) } \cos \theta = \frac{1}{2}$$



Q97. There exists a polynomial P of degree 5 with the property that if z is a complex number such that $z^5 + 2004z = 1$ then $P(z^2) = 0$. Find the value of $\left\lfloor \frac{P(1)}{P(-1)} \right\rfloor$ (where $\lfloor . \rfloor$ denote greatest integer

function)

Ans. 0

Sol. Sol. $\therefore P(z^2) = 0$ and $z^5 + 2004z = 1$
Let $z^2 = t \Rightarrow P(t) = 0$ and $z(z^4 + 2004) = 1$
 $\Rightarrow P(t) = 0$ and $(t^2 + 2004)^2 = 1$
 $\Rightarrow P(t) = 0$ and $t(t^2 + 2004)^2 - 1 = 0$
 $\Rightarrow P(t) = t(t^2 + 2004)^2 - 1$
 $\therefore \left[\frac{P(1)}{P(-1)} \right] = \left[\frac{2005^2 - 1}{2005^2 + 1} \right]$
 $\Rightarrow \left[\frac{P(1)}{P(-1)} \right]$
 $= 0$

Q98. If $z_1, z_2, z_3 \in C$ satisfying $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1 z_2 z_3 = 1$. Also $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$. Then find the value of $[|z_1 + z_2^2 + z_3^3|]$, where $[.]$ denotes the greatest integer function _____.

Ans. 2

Sol. $z_1 z_2 + z_2 z_3 + z_3 z_1$
 $= z_1 z_2 z_3 (\bar{z}_1 + \bar{z}_2 + \bar{z}_3)$
 $\Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 1$
 $\therefore z_1, z_2, z_3$ satisfy $z^3 - z^2 + z - 1 = 0$
or $z_1 = -i$
 $z_2 = 1$
 $z_3 = i \Rightarrow |z_1 + z_2^2 + z_3^3| = |-i + 1 - i| = \sqrt{5}$

Q99. If $\alpha = e^{\frac{2\pi i}{7}}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$, such that $\sum_{r=0}^6 f(\alpha^r x) = n(A_0 + A_7 x^7 + A_{14} x^{14})$, then the value of n is _____.

Ans. 7

Sol. Sol. $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k = \sum_{k=0}^{20} A_k x^k$
 $\therefore \sum_{r=0}^6 f(\alpha^r x) = f(x) + f(\alpha x) + f(\alpha^2 x) + f(\alpha^3 x) + f(\alpha^4 x) + f(\alpha^5 x) + f(\alpha^6 x)$
 $= \sum_{k=0}^{20} \left(A_k x^k + A_k (\alpha x)^k + A_k (\alpha^2 x)^k + \dots + A_k (\alpha^6 x)^k \right)$
 $= \sum_{k=0}^{20} \left\{ A_k x^k \left(1 + \alpha^k + (\alpha^2)^k + (\alpha^3)^k + \dots + (\alpha^6)^k \right) \right\} = A_0 x^0(7) + A_7 x^7(7) + A_{14} x^{14}(7)$
 $= 7[A_0 + A_7 x^7 + A_{14} x^{14}] \Rightarrow n = 7$

Q100. All the face cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed without looking at them and then randomly drawn two

cards simultaneously from the remaining. If the probability that two cards drawn are both aces

is $\frac{p \binom{38}{20}}{40C_{20} \cdot 20C_2}$, find p _____.

Ans. 6

Sol. $52 \xrightarrow{\text{face card removed}} 40 \xrightarrow{20 \text{ drawn randomly}}$

Let E_0 : 20 cards randomly removed has no aces.

E_1 : 20 cards randomly removed has exactly one ace.

E_2 : 20 cards randomly removed has exactly 2 aces.

E : event that 2 drawn from the remaining 20 cards has both the aces.

$$P(E) = P(E \cap E_0) + P(E \cap E_1) + P(E \cap E_2)$$

$$= P(E_0) \cdot P(E/E_0) + P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2)$$

$$= 40 \begin{cases} 4 \text{ aces} \\ 36 \text{ other} \end{cases}$$

$$= \frac{{}^4C_0 \cdot {}^{36}C_{20}}{40C_{20}} \cdot \frac{{}^2C_2}{{}^{20}C_2} + \frac{{}^4C_1 \cdot {}^{36}C_{19}}{40C_{20}} \cdot \frac{{}^3C_2}{{}^{20}C_2} + \frac{{}^4C_2 \cdot {}^{36}C_{18}}{40C_{20}} \cdot \frac{{}^2C_2}{{}^{20}C_2}$$

$$= \frac{{}^{36}C_{20} \cdot {}^4C_2 + {}^4C_1 \cdot {}^{36}C_{19} \cdot {}^3C_2 + {}^4C_2 \cdot {}^{36}C_{18} \cdot {}^2C_2}{40C_{20} \cdot {}^{20}C_2}$$

$$= \frac{6 \cdot {}^{36}C_{20} + 12 \cdot {}^{36}C_{19} + 6 \cdot {}^{36}C_{18}}{40C_{20} \cdot {}^{20}C_2}$$

$$= \frac{6 [{}^{36}C_{20} + {}^{36}C_{19} + {}^{36}C_{18}]}{40C_{20} \cdot {}^{20}C_2}$$

$$= \frac{6 ({}^{37}C_{20} + {}^{37}C_{19})}{40C_{20} \cdot {}^{20}C_2}$$

$$= \frac{6 \binom{38}{20}}{40C_{20} \cdot {}^{20}C_2}$$

$$\Rightarrow p = 6$$

Q101. We say that a subset S of a finite set U is large if $|S| > |U \setminus S|$. Here $U \setminus S$ denotes the elements of U which are not in S and the notation $|T|$ denotes the number of elements in a set T. Let x be the number of large subsets of the set $X = \{1, 2, \dots, 10\}$, and let y be the number of large subsets of $Y = \{1, 2, \dots, 9\}$. Which of the following is true?

Ans. -

Sol. -

Q102. The number of possible integral values of k such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y + k^2 = 0$ have exactly two common tangents is λ . A biased coin which comes up heads three times as often as tails is tossed. If it shows heads, a chip is drawn from urn-I which contains 2 white chips and 5 red chips. If the coin comes up tails, a chip is drawn from urn-II which contains λ white and 4 red chips. Given that a red chip was drawn, what is the probability that the coin came up heads? 4

Ans. -

Sol. -

Q103. Recall that if h is a function from X to Y and g is a function from Y to Z then, $g \circ h$ is the function from X to Z such that $(g \circ h)(x) = g(h(x))$, for all $x \in X$.

Let S be the set of all functions f from $\{1, 2, 3, 4, 5, 6\}$ to $\{1, 2, 3, 4, 5, 6\}$ such that $f \circ f = f$.

(a) Compute the number of functions $f \in S$ whose range has three elements.

(b) What is the cardinality of S ?

Ans. -

Sol. If y is an element in the range of f then $f(y) = y$ since f is idempotent. Because if z maps to y , $y = f(z) = f(f(z)) = f(y)$.

Can pick 3 elements in the range of f in $\binom{6}{3}$ ways. Now these elements have to go to themselves under f since $f \circ f = f$. And the remaining 3 elements have to map to one of them. So the number of such functions is

$3 * 3 * 3 = 27$. So the total number of functions with range of size 3 is $\binom{6}{3} * 27$.

Q104. The area bounded by a curve, the axis of co-ordinates & the ordinate of some point of the curve is equal to the length of the corresponding arc of the curve. If the curve passes through the point

$P(0, 1)$ then the equation of this curve will be $y = \frac{1}{k}(e^x + e^{-x})$. Let $P(z) = z^3 + az^2 + bz + c$

where $a, b, c \in \mathbb{R}$ there exists a complex number w such that the three roots of $P(z) = 0$ are $w + 3i$, $w + 9i$ and $2w - 4$ where $i^2 = -1$. Find the unit digit of the sum $|a + b + c + k|$.

Ans. -

Sol. -

Q105. If the sum of all the odd divisors of $10!$, which are of the form $3n + 1$ is S_1 and which are of the form $3n - 1$ is S_2 . Then the remainder obtained by dividing S_1 by S_2 is _____.

Ans. -

Sol. -

Q106. Match the following column :

	Column-I		Column-II
(a)	Let $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{5}{2}$ and ${}^{n+1}C_r = 21$ then ${}^nC_r - {}^nC_{r-1}$ is equal to	(p)	1
(b)	In a triangle ABC with usual notations, if $abc = (b + c - a)(c + a - b)(a + b - c)$ then $\frac{\sin \frac{A}{2} + \sin \frac{B}{2}}{\sin \frac{C}{2}}$ is equal to	(q)	2
(c)	In an isosceles triangle, whose one angle is 150° and radius of its incircle is $\sqrt{3}$ and its area is Δ then $\left(\Delta \tan^2 \frac{7\frac{1}{2}^\circ}{2} \cot 15^\circ\right)$ is equal to	(r)	6

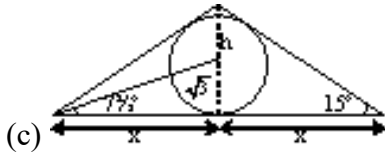
(d)	If a,b,c,d are in H.P. and a,c,d are in G.P. then $\frac{cd}{ab}$ can be equal to	(s)	9
		(t)	Odd natural number

Choose the correct option :

- (A) $a - q; b - s, t; c - t; d - p, r, t$ (B) $a - q; b - t; c - s, t; d - p, r, t$
 (C) $a - s, t; b - q; c - t; d - p, r, t$ (D) $a - q; b - t; c - p, r, t; d - s, t$

Ans. C

- Sol. (a) Let ${}^nC_r = 5x$ and ${}^nC_{r-1} = 2x^n$
 $\Rightarrow {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r = 21$
 $\Rightarrow x = 3$
 $\Rightarrow {}^nC_r - {}^nC_{r-1} = 9$
 (b) $abc = (b + c - a)(c + a - b)(a + b - c)$
 $\Rightarrow 4\Delta R = (2s - 2a)(2s - 2b)(2s - 2c)$
 $\Rightarrow 4\Delta Rs = 8s(s - a)(s - b)(s - c)$
 $\Rightarrow 4\Delta Rs = 8\Delta^2$
 $\Rightarrow Rs = 2\Delta$
 $\Rightarrow R = 2r$
 $\Rightarrow$ triangle is equilateral
 $\Rightarrow A = B = C = 60^\circ$



$$\tan\left(7\frac{1^\circ}{2}\right) = \frac{\sqrt{3}}{x}$$

$$x = \sqrt{3} \cot 7.5^\circ$$

$$2x = 2\sqrt{3} \cot 7.5^\circ$$

$$\tan 15^\circ = \frac{h}{x}$$

$$h = x \tan 15^\circ$$

$$h = \sqrt{3} \cot\left(7\frac{1^\circ}{2}\right) \tan 15^\circ$$

$$\Delta = \frac{1}{2} \cdot 2\sqrt{3} \cot^2\left(7\frac{1^\circ}{2}\right) \sqrt{3} \cdot \tan 15^\circ$$

$$\Delta \equiv \cot^2\left(7\frac{1^\circ}{2}\right) \tan 15^\circ$$

(d) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d} \Rightarrow A.P$
 Let $\frac{1}{a} = A - 3D$
 $\frac{1}{b} = A - D$
 $\frac{1}{c} = A + D$

$$\frac{1}{d} = A + 3D$$

$$\Rightarrow \frac{cd}{ab} = 1 \text{ when } D = 0$$

or when $A = -5D$

then $\frac{cd}{ab} = 6$

a, c, d are in G.P,

$$\Rightarrow c^2 = ad$$

$$\Rightarrow (A + D)^2 = A^2 - 9D^2$$

$$\Rightarrow 2Ad + 10D^2 = 0$$

$$\Rightarrow D = 0 \text{ or } A = -5D$$

Q107. Match the following column :

	Column-I		Column-II
(a)	Out of four machines, exactly two are faulty, they are tested one by one in a random order till both faulty machines are identified, then the probability that only two test are needed.	(p)	$\frac{1}{2}$
(b)	A die with 6 faces marked 1, 1, 4, 3, 3, 3 is tossed twice. Find the probability of getting sum 4.	(q)	$\frac{1}{4}$
(c)	$(a_1, b_1, c_1), (a_2, b_2, c_2)$ and (a_3, b_3, c_3) are direction ratios of three perpendicular lines and direction ratio of line equally inclined to them is given by $k(a_1 + a_2 + a_3), k(b_1 + b_2 + b_3), k(c_1 + c_2 + c_3)$. Then, k is given by	(r)	$\frac{1}{3}$
(d)	If three points are lying in a plane what is the probability that a triangle will be formed by joining them	(s)	$\frac{1}{6}$

Choose the correct option :

- (A) a-r, b-s, c-p, d-q (B) a-q, b-r, c-s, d-p (C) a-r, b-r, c-q, d-p (D) a-p, b-q, c-r, d-s

Ans. C

Sol. (a) P (two faulty identified) + P (two correct identified)

$$= 2 \left(\frac{{}^2C_2}{{}^4C_2} \right)$$

$$= 2 \left(\frac{1 \cdot 2}{4 \cdot 3} \right)$$

$$= 2 \left(\frac{2}{12} \right) = \frac{1}{3}$$

$$(b) n(S) = 6 \times 6 = 36$$

$$n(E) = 12$$

$$P = \frac{12}{36}$$

$$= \frac{1}{3}$$

(c) k may be any real constant.

(d) $n(E) = 1, n(S) = 2$

Q108. Match the following column :

Column-I		Column-II	
(a)	If the binomial coefficients of the r^{th} , $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ term in the expansion of $(1+x)^{14}$ are in A.P, then r is equal to	(p)	5
(b)	The sum of the coefficients in the polynomial expansion of $\{[1+x+x^2+x^3+x^4+\dots+(x)^{n-1}]\}^m$, $(n, m \in N)$, is	(q)	4
(c)	Sum of the series $\sum_{r=1}^n (-1)^r \cdot {}^nC_r [1+i^r+i^{2r}+\dots+i^{(n-1)r}]$ where $(n=4k, k \in I)$ is equal to (where $i = \sqrt{-1}$)	(r)	0
(d)	If $3^{37} = 80\lambda + k$, where $\lambda \in N$, then k is	(s)	3

Choose the correct option :

- (A) a-p; b-q; c-r; d-s (B) a-r; b-p c-q; d-s (C) a-p; b-q; c-s; d-r (D) a-p; b-r; c-r; d-s

Ans. D

Sol. (a) We are given that $2 {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1}$
 $\Rightarrow {}^{14}C_r = 2(r+1)(15-r) = r(r+1) + (15-r)(14-r)$
or $30 + 28r - 2r^2 = r^2 + r + 210 - 29r + r^2$
or $r^2 - 14r + 45 = 0 \Rightarrow (r-5)(r-9) = 0$
(b) $[\{1+x+x^2+x^3+x^4+\dots+(x)^{n-1}\}(1-x)]^m = \left[\frac{1-(x)^n}{1-x} (1-x) \right]^m = (1-(x)^n)^m$
 $= {}^mC_0 + {}^mC_1(-x)^n + \dots + (-1)^{mm} {}^mC_m (X)^{mn}$ So sum of all the coefficients is 0.
(c) $\sum_{r=1}^n (-1)^r \cdot {}^nC_r \left(\frac{1-i^{nr}}{1-i^r} \right)$
 $\sum_{r=1}^n (-1)^r \cdot {}^nC_r \left(\frac{1-i^{4kr}}{1-i^r} \right) = \sum_{r=1}^n (-1)^r \cdot {}^nC_r \left(\frac{1-1}{1-i^r} \right)$
 $= 0$
(d) $3^{37} = 3^{4 \times 9} \cdot 3 = 3(81)^9 = 3(80+1)^9$
 $= 3({}^9C_0 80^0 + {}^9C_1 80^1 + \dots + {}^9C_9)$
hence remainder is 3

Q109. Five balls are to be placed in three boxes. Each can hold all the five balls. In how many different ways can we place the ball so that no box remains empty, if
Match the following column :

Column-I	Column-II
(a) Balls and boxes are all different	(p) 2
(b) Balls are identical but boxes are different	(q) 25
(c) Balls are different but boxes are identical	(r) 150
(d) Balls and boxes are identical	(s) 6

Choose the correct option :

- (A) $a - r, b - s; c - p; d - q$ (B) $a - r; b - p; c - s; d - q$ (C) $a - r; b - s; c - q; d - p$
 (D) $a - p; b - r; c - s; d - q$

Ans. C

- Sol. (a) $35 - ({}^3C_2 2^5 - {}^3C_1 1^5) = 150$
 (b) $x_1 + x_2 + x_3 = 5, x_1 \geq 1, x_2 \geq 1, x_3 \geq 1$
 $\Rightarrow {}^{2+3-1}C_{3-1} = 6$
 (c) $\frac{5!}{3!(1!)^2} \frac{1}{2!} + \frac{5!}{(2!)^2(1!)^2} \frac{1}{2!} = 25$
 (d) Only 2 cases i.e. (3, 1, 1) and (2, 2, 1)